Algebra 1A202425 - lecture notes

Algebra 1A (University of Bath)

Mathematical Sciences: Algebra 1A

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Contents

1 Numbers and Arithmetic

1.1 Notation

Let's start at the very beginning with some familiar sets of numbers:

- the *natural numbers* $\mathbb{N} = \{1, 2, \dots\}$. Here we choose[1] to exclude zero;
- the *integers* $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$;
- the *non-negative integers* $\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$, sometimes called *whole numbers*;
- the *rationals* $\mathbb{Q} = \left\{\frac{m}{n} \mid m, n \in \mathbb{Z} \text{ with } n \neq 0\right\}$, where $\frac{m}{n} = \frac{p}{q} \iff mq = np$. In this definition, the vertical bar is read as "such that".
- the *real numbers* $\mathbb{R}$ (defined in Analysis 1A and 'Sequences and functions').

If we want to work with an arbitrary integer, say, we write 'Let $n \in \mathbb{Z}$.' and we read this as '*Let n be an integer.*', or equivalently, '*Let n be an element of the set of integers.*'.

1.2 Complex numbers

We first introduce an additional set of numbers. As you know well, for $a, b, c \in \mathbb{R}$ with $a \neq 0$, solutions of $ax^2 + bx + c = 0$ are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

but this can be problematic because $b^2 - 4ac$ might be negative. We can ensure that every quadratic equation has at least one solution by introducing *complex numbers*. We'll see that every real number has at least one square root: for real numbers of the form $\lambda \geq 0$, the solutions to $x^2 = \lambda$ are $x = \pm\sqrt{\lambda}$, and the solutions to $x^2 = -\lambda$ are $x = \pm i\sqrt{\lambda}$ where i is the *imaginary number* defined below.

Definition 1.1 (**Complex numbers**)**.** The set of *complex numbers* is

$$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\},$$

where $x + iy = a + ib \iff x = a$ and $y = b$. For the complex number $z = x + iy$, we call $x = \Re(z)$ the *real part* of z, and $y = \Im(z)$ the *imaginary part* of z. The operations of *sum* and *product* of complex numbers are defined by:

$$\begin{aligned} (a + ib) + (x + iy) &= (a + x) + i(b + y) \\ (a + ib)(x + iy) &= ax + i(bx + ay) + i^2 by = (ax - by) + i(bx + ay). \end{aligned}$$

This includes the special case $a(x + iy) = ax + iay$, for $a \in \mathbb{R}$.

[1] There is no universal agreement in mathematics as to whether zero should lie in $\mathbb{N}$. In some contexts, it's more convenient to include zero, while in others, it's better to exclude zero. In Year 1 Mathematics in Bath, we have chosen to exclude 0 from the set $\mathbb{N}$.

Example 1.2. The rule for multiplication for $a = x = 0$ and $b = y = 1$ shows that the imaginary number $i \in \mathbb{C}$ satisfies $i^2 = -1$. It is therefore often referred to as 'the square root of minus one', though we'll see that it's better to say 'a square root of minus one' (because there are two such square roots in $\mathbb{C}$, namely i and $-i$).

The complex numbers inherit many familiar properties from the real numbers[2] .

Proposition 1.3 (Properties of complex numbers). *Addition of complex numbers satisfies the following:*

F1 $\forall v, w, z \in \mathbb{C}$, *we have* $(v + w) + z = v + (w + z)$.

F2 *The number* $0 := 0 + i0 \in \mathbb{C}$ *satisfies:* $\forall z \in \mathbb{C}$, $0 + z = z = z + 0$.

F3 *For* $z = x + iy \in \mathbb{C}$, *the number* $-z := -x + i(-y)$ *satisfies* $z + (-z) = 0 = -z + z$.

F4 $\forall w, z \in \mathbb{C}$, *we have* $w + z = z + w$.

Addition and multiplication are compatible, in the sense that $\cdot$ *is distributive over* $+$*:*

F5 $\forall a, w, z \in \mathbb{C}$, $a \cdot (w + z) = a \cdot w + a \cdot z$ *and* $(w + z) \cdot a = w \cdot a + z \cdot a$.

Multiplication of complex numbers satisfies the following:

F6 $\forall v, w, z \in \mathbb{C}$, *we have* $(v \cdot w) \cdot z = v \cdot (w \cdot z)$.

F7 *The number* $1 := 1 + i0 \in \mathbb{C}$ *satisfies:* $\forall z \in \mathbb{C}$, $1 \cdot z = z = z \cdot 1$.

F8 $\forall w, z \in \mathbb{C}$, *we have* $w \cdot z = z \cdot w$.

F9 $\forall z \in \mathbb{C}$ *with* $z \neq 0$, *there is* $z^{-1} \in \mathbb{C}$ *satisfying* $z^{-1} \cdot z = 1 = z \cdot z^{-1}$.

Proof. Most of these properties follow by writing each complex number in the form $x+iy$, and deducing the result from the correpsonding statement for real numbers. For example, for **F4**, write $z = x + iy$ and $w = a + ib$ and compute

$$w + z = (a + x) + i(b + y) = (x + a) + i(y + b) = z + w,$$

where the second equality holds because addition of real numbers is commutative, i.e. the property **F4** holds for real numbers. I urge you to write out the proofs for yourself. (Like most things in life, you learn mathematics best by doing it, not merely reading about it.)

Property **F9**, which says that every nonzero complex number has a multiplicative inverse, is the least obvious so let's prove it. Suppose $z = x + iy$ is nonzero; this is equivalent to saying that at least one of x, y is nonzero. Then the complex number

$$z^{-1} := \frac{x - iy}{x^2 + y^2} = \frac{x}{x^2 + y^2} + \frac{-y}{x^2 + y^2} i$$

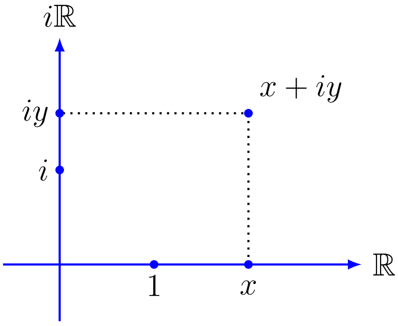
[2] A set with sum and product operations that satisfies certain natural properties (like 'addition and multiplication are both associative', etc) is called a *ring*; in fact, $\mathbb{C}$ is a *field* because it satisfies all of **F1** to **F9**; see Section 3.1.

satisfies

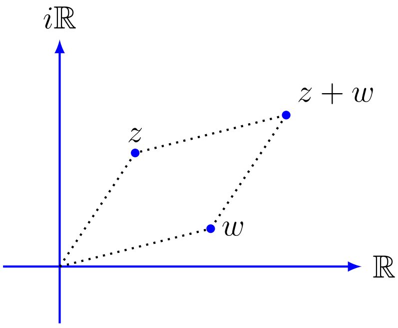
$$z \cdot z^{-1} = z^{-1} \cdot z = (x+iy)\frac{(x-iy)}{x^2+y^2} = \frac{x^2+y^2}{x^2+y^2} = 1$$

as required. □

Complex numbers can be interpreted geometrically in the *(Argand) plane* by plotting the number $z = x + iy$ as the ordered pair (x, y) as shown above. Clearly, specifying both x and y uniquely determines the complex number $x + iy$, and vice versa. In the Argand plane, we refer to the x- and y- axes as the *real* and *imaginary* axes $\mathbb{R}$ and $i\mathbb{R}$ respectively.



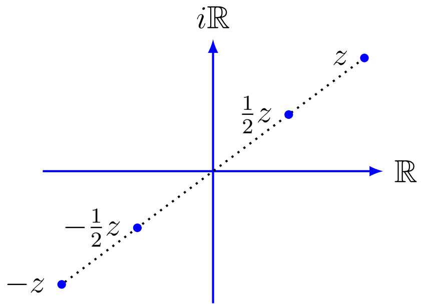
This allows for a geometric interpretation of many natural operations on complex numbers. For example, given $z = x + iy$, adding the number $w = a + ib$ corresponds to translation by $(a, b) \in \mathbb{R}^2$ as shown:



On the other hand, multiplication by $a \in \mathbb{R}$ is scaling if $a > 0$, or scaling with reflection in the origin (rotation by π radians) if $a < 0$.

Definition 1.4 (**Complex conjugate and modulus**)**.** Let $z = x + iy \in \mathbb{C}$. Then $\overline{z} := x - iy$ is called the *complex conjugate* (reflection in the real axis) of z, and $|z| := \sqrt{x^2 + y^2}$ is called the *modulus* (or magnitude) of z.

Remarks 1.5. 1. The complex conjugate can be used to simplify the presentation of a complex number that is written as a fraction involving two complex numbers. For example, we can rewrite $(1 + 2i)/(3 + 4i)$ by multiplying by 1 (of course!), but



we choose to regard 1 as the quotient of $\overline{3+4i} = 3-4i$ by itself, and simplify the result, namely:

$$\frac{1+2i}{3+4i} = \frac{1+2i}{3+4i} \cdot 1 = \frac{(1+2i)}{(3+4i)}\frac{(3-4i)}{(3-4i)} = \frac{(3+8)+(6-4)i}{9+16} = \frac{11}{25} + \frac{2}{25}i.$$

2. Observe that $|z| \geq 0$, with equality if and only if $x = y = 0$, i.e. $z = 0$. Also $|z_1 - z_2|$ is the distance from z_1 to z_2 in the plane.

Lemma 1.6. *For $z, z_1, z_2 \in \mathbb{C}$ we have*

1. $\Re(z) = \dfrac{z+\overline{z}}{2}, \quad \Im(z) = \dfrac{z-\overline{z}}{2i}$
2. $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}, \quad \overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$
3. $|z|^2 = z \cdot \overline{z}$
4. $|z_1 z_2| = |z_1| \cdot |z_2|, \quad |z^n| = |z|^n$ *for any* $n \in \mathbb{N}$.

Proof. We leave (1) and the first part of (2) as simple exercises. For the second part of (2), let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. Then

$$\begin{aligned}
\overline{z_1 \cdot z_2} &= \overline{(x_1+iy_1)(x_2+iy_2)} \\
&= \overline{x_1x_2 - y_1y_2 + i(x_1y_2 + y_1x_2)} \\
&= x_1x_2 - y_1y_2 - i(x_1y_2 + y_1x_2)
\end{aligned}$$

which we need only compare with

$$\overline{z_1} \cdot \overline{z_2} = (x_1 - iy_1)(x_2 - iy_2) = (x_1x_2 - y_1y_2) - i(x_1y_2 + y_1x_2).$$

For (3), write $z = x + iy$. Then $z \cdot \overline{z} = (x+iy)(x-iy) = x^2 - (iy)^2 = x^2 + y^2 = |z|^2$. For the first statement in (4), compute

$$\begin{aligned}
|z_1z_2|^2 &= (z_1z_2)\overline{(z_1z_2)} && \text{by (3)} \\
&= z_1 \cdot z_2 \cdot \overline{z_1} \cdot \overline{z_2} && \text{by (2)} \\
&= z_1 \cdot \overline{z_1} \cdot z_2 \cdot \overline{z_2} \\
&= |z_1|^2|z_2|^2 && \text{by (3).}
\end{aligned}$$

Now take the radical (=positive square root) of both sides. To prove the second part of (4), we use induction. The base case is $n = 1$, where the statement is obvious. Suppose that the statement is true for a given $n \in \mathbb{N}$, so $|z^n| = |z|^n$. Apply the first part of (4) to the product $z \cdot z^n$ to see that

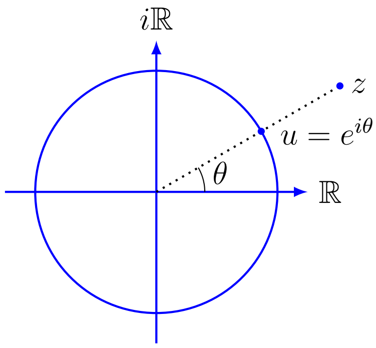
$$\begin{aligned} |z^{n+1}| = |z \cdot z^n| &= |z| \cdot |z^n| && \text{by the first statement in (4),} \\ &= |z| \cdot |z|^n && \text{by the inductive assumption for } n, \\ &= |z|^{n+1}, \end{aligned}$$

so the result holds for $n+1$. Thus, the result holds for all $n \in \mathbb{N}$ by induction. □

Our next goal is to introduce the convenient 'polar form' for a complex number. To this end, note that for any nonzero $z \in \mathbb{C}$, the complex number $u = z/|z|$ satisfies

$$|u|^2 = u\overline{u} = \frac{z\overline{z}}{|z|^2} = 1,$$

so $|u| = 1$. In fact, u is the intersection of the unit circle $S^1 = \{w \in \mathbb{C} \mid |w| = 1\}$ with the straight line from the origin that passes through z.



For a nonzero $z \in \mathbb{C}$, the *argument* of z is the angle $\theta = \arg(z)$ subtended by the unique straight line through the origin and z, taken to lie in the interval $[0, 2\pi)$ as shown. For $0 \leq \theta < 2\pi$, the unique point on the unit circle with argument θ (denoted u above) has real and imaginary parts given by $\cos\theta$ and $\sin\theta$, so $u = \cos\theta + i\sin\theta$. One of the key results in Analysis 1B is to prove *Euler's formula*[3], namely that

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

You may take this purely as the definition of $e^{i\theta}$ for now, i.e. when we write $e^{i\theta}$ in Algebra, we simply mean the complex number $\cos\theta + i\sin\theta$ as shown in the diagram above.

Definition 1.7 (**Polar form**)**.** The *polar form* of any nonzero $z \in \mathbb{C}$ is

$$z = re^{i\theta} = r(\cos\theta + i\sin\theta),$$

where $r = |z|$ is the modulus of z and $\theta = \arg(z)$.

[3]The special case when $\theta = \pi$ shows that $e^{i\pi} = -1$, leading to the beautiful equation $e^{i\pi} + 1 = 0$ that is often referred to as *Euler's identity*.

Example 1.8. For $z = 1 + i$, we have $|z| = \sqrt{2}$ and $\arg(z) = \pi/4$; for $z = i$, we have $|z| = 1$ and $\arg(z) = \pi/2$; and for $z = -1$, we have $|z| = 1$ and $\arg(z) = \pi$.

Remark 1.9. Since cos and sin are both periodic with period 2π, we have $e^{i\theta} = e^{i(\theta+2\pi k)}$ for each $k \in \mathbb{Z}$.

Proposition 1.10 (The unit circle is a group). *The unit circle $S^1 = \{w \in \mathbb{C} \mid |w| = 1\}$ is closed under complex multiplication; more explicitly,*

$$e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1+\theta_2)}.$$

Proof. If $u_1, u_2 \in S^1$, then $|u_1 u_2| = |u_1| \cdot |u_2| = 1 \cdot 1 = 1$, so S^1 is 'closed under multiplication' (meaning that the product of two complex numbers that lie on the unit circle is a complex number on the unit circle). Furthermore, the angle sum formulae give

$$\begin{aligned}
&(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) \\
&\qquad = (\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2) \\
&\qquad = \cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2).
\end{aligned}$$

which is simply a restatement of the desired equality $e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$. □

More generally, rather than just multiply complex numbers on the unit circle, let $u = e^{i\theta}$ lie on the unit circle and let $z = se^{i\mu}$ be an arbitrary nonzero complex number. Then multiplication by u is

$$z \mapsto u \cdot z = e^{i\theta} \cdot se^{i\mu} = se^{i\theta} \cdot e^{i\mu} = se^{i(\theta+\mu)},$$

which rotates z by the angle $\theta = \arg(u)$ about the origin. More generally still, for any nonzero $w = re^{i\theta} \in \mathbb{C}$, multiplication by w is

$$z \mapsto w \cdot z = re^{i\theta} \cdot se^{i\mu} = rse^{i\theta} \cdot e^{i\mu} = rse^{i(\theta+\mu)},$$

which is the composition of rotation by $\arg(w) = \theta$ and scaling by $|w| = r$.

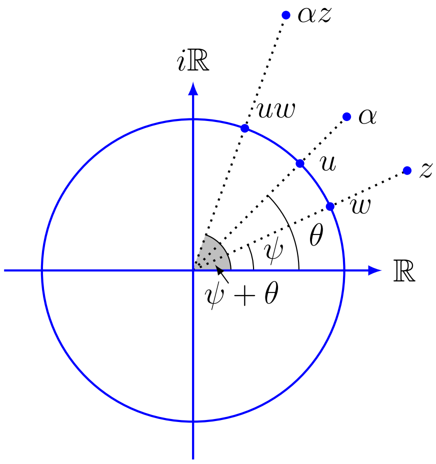
Corollary 1.11 (de Moivre's Theorem). *For $n \in \mathbb{Z}$, we have $\left(e^{i\theta}\right)^n = e^{in\theta}$, i.e.*

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta.$$

Proof. First, we prove the result for all $n \in \mathbb{N}$ by induction. The result is clear for $n = 1$. As our inductive hypothesis, suppose that the statement holds for some $k \in \mathbb{N}$. Then

$$\begin{aligned}
\left(e^{i\theta}\right)^{k+1} &= \left(e^{i\theta}\right)^k \cdot e^{i\theta} \\
&= e^{ik\theta} \cdot e^{i\theta} && \text{by the inductive hypothesis} \\
&= e^{i(k+1)\theta} && \text{by Proposition 1.10,}
\end{aligned}$$

so the statement holds for $k + 1$. It follows by induction that it holds for all $n \in \mathbb{N}$.



Next, we note that the result holds for $n = 0$, simply because

$$(\cos\theta + i\sin\theta)^0 = 1 = \cos 0 + i\sin 0.$$

It remains to prove the result for the negative integers, and again we proceed by (descending) induction. For $n = -1$, Proposition 1.10 gives

$$e^{i\theta} \cdot e^{-i\theta} = e^{i\cdot 0} = \cos 0 + i\sin 0 = 1,$$

so $\left(e^{i\theta}\right)^{-1} = e^{-i\theta}$ as required. For the inductive step, suppose that the statement holds for some integer $-k$ where $k \in \mathbb{N}$. Then

$$\begin{aligned}
\left(e^{i\theta}\right)^{-k-1} &= \left(e^{i\theta}\right)^{-k} \cdot e^{-i\theta} & \\
&= e^{-ik\theta} \cdot e^{-i\theta} & \text{by the inductive hypothesis} \\
&= e^{i(-k-1)\theta} & \text{by Proposition 1.10,}
\end{aligned}$$

so the statement holds for $-(k+1)$. It follows by (descending) induction that it holds for all negative integers. $\square$

Definition 1.12 (Roots of unity). Let $n \in \mathbb{N}$. An *nth root of unity* is any complex number $z \in \mathbb{C}$ that satisfies $z^n = 1$.

Proposition 1.13. *The nth roots of unity are precisely $z = e^{2\pi ik/n}$ for $0 \le k \le n-1$. In particular,*

$$z^n - 1 = \prod_{k=0}^{n-1} \left(z - e^{2\pi ik/n}\right).$$

Proof. We claim first that if z is an nth root of unity, then $|z| = 1$. To establish this, we will use the equality $|z^n| = |z|^n$ from Lemma 1.6 (4). Indeed, if $|z| > 1$, then $|z^n| = |z|^n > 1$ and if $|z| < 1$, then $|z^n| < 1$; in either case, we could not have $z^n = 1$. This proves the claim, so each root of unity z lies on the unit circle, and hence $z = e^{i\theta}$ for some $0 \le \theta < 2\pi$.

Now apply de Moivre's Theorem to see that

$$e^{i0} = 1 = z^n = \left(e^{i\theta}\right)^n = e^{in\theta}.$$

Comparing the exponents shows that the possible values of θ satisfy

$$n\theta = 0 + 2\pi k \quad \text{for some } k \in \mathbb{Z}.$$

However, $\theta \in [0, 2\pi)$, so multiplying by n forces $n\theta \in [0, 2\pi n)$. The previous line shows that $n\theta$ must be an integer-multiple of 2π, and the only such values in $[0, 2\pi n)$ are $0, 2\pi, 4\pi, \dots, 2\pi(n-1)$. Now divide by n to see that the only possible values of θ are

$$\theta \in \left\{0, \frac{2\pi}{n}, \frac{4\pi}{n}, \dots, \frac{2\pi(n-1)}{n}\right\}$$

which is what we want to prove. $\square$

Notice that $z^n - 1$, a polynomial of degree n, has n roots in $\mathbb{C}$. This fact is very general, and has far-reaching consequences across mathematics.

Theorem 1.14 (**Fundamental Theorem of Algebra**). *Every polynomial $f(z)$ in one variable with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$, that is, a solution to $f(z) = 0$.*

Corollary 1.15. *Every polynomial of degree n with coefficients in $\mathbb{C}$ has precisely n roots in $\mathbb{C}$ if we count roots 'with multiplicity', i.e. each is a product of n linear factors: if*

$$f(z) = a_n z^n + \ldots + a_1 z + a_0,$$

with $a_j \in \mathbb{C}$ and $a_n \neq 0$, then $f(z) = a_n \prod_{i=1}^{n}(z - \alpha_i)$, for some $\alpha_i \in \mathbb{C}$.

Definition 1.16 (**Order of a root of unity**). Let $z \in \mathbb{C}$ be a root of unity, i.e. $z^n = 1$ for some $n \in \mathbb{N}$. The *order* of z, denoted $\operatorname{ord}(z)$, is the smallest natural number m such that $z^m = 1$. If $\operatorname{ord}(z) = m$, we say that z is a *primitive* mth root of unity.

Example 1.17. The sixth roots of unity are $e^{2\pi ik/6} = e^{i\pi k/3}$ for $k = 0, 1, \dots, 5$. To find the order of each root, compute powers of the exponents $\pi k/3$ for $k = 0, 1, \dots, 5$ and check if they are multiples of 2π. For $k = 0, 1, 2$, we find that

$$\operatorname{ord}(1) = 1; \quad \operatorname{ord}(e^{i\pi/3}) = 6; \quad \operatorname{ord}(e^{2\pi i/3}) = 3;$$

and for $k = 3, 4, 5$, we compute

$$\operatorname{ord}(-1) = 2; \ \operatorname{ord}(e^{-2\pi i/3}) = 3; \quad \operatorname{ord}(e^{-\pi i/3}) = 6.$$

The primitive roots of the same order, give rise to the factors in the following factorisation:

$$z^6 - 1 = (z^3 - 1)(z^3 + 1) = (z - 1)(z^2 + z + 1)(z + 1)(z^2 - z + 1)$$

since $z^2 + z + 1 = (z - e^{i2\pi/3})(z - e^{-i2\pi/3})$ and $z^2 - z + 1 = (z - e^{i\pi/3})(z - e^{-i\pi/3})$.

Remark 1.18. In the exercises, you'll show that the primitive nth roots of unity are precisely the numbers of the form $e^{2\pi ik/n}$, where $k \in \{0, 1, \dots, n-1\}$ is such that k and n have no common factors (other than ± 1).

1.3 Prime factorisation

Now, let's leave the complex numbers behind and instead work with the set of numbers that are most familiar: the set $\mathbb{Z}$ of integers. The key is to study each integer in terms of its prime divisors (= prime factors).

Definition 1.19 (Prime numbers). Let $a, b \in \mathbb{Z}$. We say that a *divides* b, written $a \mid b$, if there exists $n \in \mathbb{Z}$ such that $b = an$. Given a positive integer $p \in \mathbb{N}$, we say that p is *prime* if $p \geq 2$ and if any $a \in \mathbb{N}$ such that $a \mid p$ must be either $a = 1$ or $a = p$.

Remarks 1.20.

1. The statement 'any $a \in \mathbb{N}$ such that $a \mid p$ must be either $a = 1$ or $a = p$. ' is written more efficiently as '$\forall a \in \mathbb{N},\ a \mid p \implies a = 1$ or $a = p$'.
2. If $m, n \in \mathbb{N}$ satisfy $m \mid n$, then $m \leq n$.
3. For all $a \in \mathbb{Z}$, $1 \mid a$ and $a \mid 0$ (because $a = 1 \cdot a$ and $0 = 0 \cdot a$). However, if $0 \mid a$, then $a = 0$, and if $a \mid 1$, then $a = \pm 1$.

Let's now formulate and prove some statements that need a little more justification.

Lemma 1.21. *Let $a, b, m \in \mathbb{Z}$.*

(1) If $a \mid b$ and $b \mid a$, then $a = \pm b$.

(2) If $m \mid a$ and $m \mid b$, then $m \mid \lambda a + \mu b$ for all $\lambda, \mu \in \mathbb{Z}$.

Proof. For (1), there exists $x, y \in \mathbb{Z}$ such that $ax = b$ and $by = a$. Substituting the first equation into the second gives $axy = a$. There are two cases: if $a \neq 0$, then cancel a to leave $xy = 1$, so $x, y = \pm 1$; otherwise, $a = 0$, in which case $b = 0 \cdot x = 0 = a$ as required.

For (2), there exists $x, y \in \mathbb{Z}$ such that $mx = a$ and $my = b$. Then for any $\lambda, \mu \in \mathbb{Z}$, we have $\lambda a + \mu b = m(\lambda x + \mu y)$, so $m \mid \lambda a + \mu b$. □

Proposition 1.22 (Product of primes). *Every integer that's greater than one is a product of primes, that is, each $m \in \mathbb{Z}$ satisfying $m \geq 2$ can be written as*

$$m = \prod_{i=1}^{k} p_i = p_1 p_2 \cdots p_k$$

for some $k \geq 1$, where $p_1, \ldots, p_k$ are primes that need not be distinct.

Remark 1.23. We present a stronger statement in Theorem 1.33 below.

Proof of Proposition 1.22. We prove the statement by induction on $m \geq 2$. For $m = 2$, the statement is true because 2 is prime. Assume by induction that the statement is true for all integers from 2 to $m - 1$. To investigate the result for m, we consider two cases. First, if m is prime, then our decomposition consists purely of m itself, so $k = 1$ and there is nothing to prove. Otherwise, m is not prime. If we negate the definition of what it means for m to be prime, we see that there exists $a \in \mathbb{N}$ satisfying $a \mid m$ where $a \neq 1$ and $a \neq m$. It follows that there must exist $b \in \mathbb{N}$ such that $m = ab$ with $1 < a, b < m$. Our inductive hypothesis gives a decomposition of both a and b into products of primes, and the product of these two decompositions is the desired decomposition of m. □

Theorem 1.24 (**Euclid**). *There are infinitely many primes.*

Proof. We prove this result by contradiction. Suppose that there are only finitely primes, say $p_1, \ldots, p_k$ for some $k \in \mathbb{N}$. Consider the integer

$$n := p_1 \cdots p_k + 1.$$

Since $n \geq 2$, Proposition 1.22 gives that n is a product of primes and hence $n = p_j m$ for some $1 \leq j \leq k$ and $m \in \mathbb{N}$. But then

$$1 = n - p_1 \cdots p_k = p_j m - p_1 \cdots p_j \cdots p_k$$

is divisible by p_j, so $p_j = 1$ which is a contradiction. Therefore, there are infinitely many primes after all. $\square$

Definition 1.25 (**lcm and gcd**). Let $m, n \in \mathbb{N}$. We say that $a \in \mathbb{N}$ is:

- a *common multiple* of m and n if $m \mid a$ and $n \mid a$. The *least common multiple of m and n*, denoted $\operatorname{lcm}(m, n)$, is the smallest such $a \in \mathbb{N}$;
- a *common divisor* of m and n if $a \mid m$ and $a \mid n$. The *greatest common divisor of m and n*, denoted $\gcd(m, n)$, is the largest such $a \in \mathbb{N}$.

We say that m and n are *coprime* if $\gcd(m, n) = 1$.

Remark 1.26. Let $g := \gcd(m, n)$. We claim that $\frac{m}{g}$ and $\frac{n}{g}$ are coprime. Indeed, let $d = \gcd(\frac{m}{g}, \frac{n}{g})$. Then $d \mid \frac{m}{g}$ and $d \mid \frac{n}{g}$, say $\frac{m}{g} = d\alpha$ and $\frac{n}{g} = d\beta$ for some $\alpha, \beta \in \mathbb{N}$. Then $m = dg\alpha$ and $n = dg\beta$, so dg divides both m and n. But we know that $g = \gcd(m, n)$, so dg dividing both m and n forces $d = 1$.

Example 1.27. As a reality check, we have that $\gcd(16, 20) = 4$ and $\operatorname{lcm}(16, 20) = 80$. Notice that $\gcd(16/4, 20/4) = \gcd(4, 5) = 1$.

Proposition 1.28 (**Bézout identity**). *For $a, b \in \mathbb{N}$, let g be the smallest positive integer in the set*

$$S = \{\lambda a + \mu b \mid \lambda, \mu \in \mathbb{Z}\}.$$

Then $g = \gcd(a, b)$. In particular, $\gcd(a, b) = \lambda a + \mu b$ for some $\lambda, \mu \in \mathbb{Z}$.

Proof. If $m \in \mathbb{N}$ is a common divisor of a and b, then $m \mid a$ and $m \mid b$, and hence $m \mid g$ by Lemma 1.21(2) because $g \in S$. Thus, if g itself divides both a and b, then it's the greatest common divisor as required[4].

[4] Following an excellent question after class, let's justify the assertion here that $g \geq m$. We know that $g, m \in \mathbb{N}$ and that $m \mid g$. Since $m \mid g$, there exists $n \in \mathbb{Z}$ satisfying $g = nm$. We know both g and m are strictly positive, so n must also be strictly positive. It's an integer, so in fact $n \in \mathbb{N}$, and in particular, $n \geq 1$. It now follows that $g = nm \geq m$ as claimed.

To see this, we may write $g = \lambda a + \mu b$ for some $a, b \in \mathbb{Z}$ because $g \in S$. Compute the 'division with remainder' of a by g to obtain $q, r \in \mathbb{Z}$ such that $a = qg + r$, where the remainder r satisfies $0 \leq r < g$. Notice now that

$$r = a - qg = (1 - \lambda q)a - (\mu q)b \in S.$$

However, g is the smallest positive element in S, so we must have $r = 0$, giving $a = qg$. This shows that $g \mid a$, and one can show similarly that $g \mid b$. Thus, g is indeed a common divisor of a and b, so by the first paragraph of the proof, we have $g = \gcd(a, b)$. The final statement of the proposition follows from the fact that $\gcd(a, b) = g \in S$. □

Algorithm 1.29 (**Euclid's algorithm**). For $a, b \in \mathbb{N}$, compute $\gcd(a, b)$ as follows:

(0) If $a = b$, then $\gcd(a, b) = a$. Otherwise, $a \neq b$, and we may assume without loss of generality that $a > b$. Set $a_0 := a$ and $a_1 := b$

(1) For $j \geq 1$, calculate the quotient with remainder of a_{j-1} by a_j to obtain $q_j, r_j \in \mathbb{N}$ satisfying
$$a_{j-1} = q_j a_j + r_j \quad \text{where } 0 \leq r_j < a_j.$$

(2) If j is the index such that $r_j = 0$, then $a_j = \gcd(a, b)$. Otherwise, define $a_{j+1} := r_j$ and increase the index j by one. Return to step (1) with this new index j.

Proof of correctness. First, observe that the algorithm terminates, because it produce a descreasing sequence $a_0 > a_1 > a_2 > \dots$ of positive integers (recall $a_{j+1} := r_j \geq 0$) which necessarily terminates in at most a_0 steps.

To see that the algorithm calculates $\gcd(a, b)$, note that $a_2 = a_0 - q_1 a_1$. Lemma 1.21(2) implies that any number dividing both a_0 and a_1 also divides both a_1 and a_2, and similarly, any number that divides a_1 and a_2 also divides a_1 and a_0. Thus, $\gcd(a_0, a_1) = \gcd(a_1, a_2)$. This is true for each j, so if n is the index for which $r_n = 0$, then $a_{n-1} = q_n a_n$, and hence

$$\gcd(a, b) = \gcd(a_0, a_1) = \gcd(a_1, a_2) = \cdots = \gcd(a_{n-1}, a_n) = a_n.$$

Thus, for the index n satisfying $r_n = 0$, we have $a_n = \gcd(a, b)$ as claimed. □

Remark 1.30. The Bézout identity result means that there exist integers x, y such that

$$\gcd(a, b) = xa + yb.$$

To write down such an equation explicitly, we're looking for a point (x, y) that lies on the straight line
$$\gcd(a, b) = xa + yb$$
(remember that $\gcd(a, b)$ is a natural number!) such that both x, y are integers. We can find the gcd and then find such a point by working backwards from Euclid's equations.

Example 1.31. To find $\gcd(20, 14)$, the algorithm produces the following equations:

$$\begin{aligned} \underline{20} &= 1 \cdot \underline{14} + \underline{6} \\ \underline{14} &= 2 \cdot \underline{6} + \underline{2} \\ \underline{6} &= 3 \cdot \underline{2} + 0. \end{aligned}$$

Therefore $2 = \gcd(20, 14)$. By working backwards, we compute the Bézout identity

$$\begin{aligned} \underline{2} &= -2 \cdot \underline{6} + \underline{14} \\ &= -2 \cdot (-\underline{14} + \underline{20}) + \underline{14} \\ &= 3 \cdot \underline{14} + (-2) \cdot \underline{20}. \end{aligned}$$

as required. This means that the point $(x, y) = (3, -2)$ lies on the line $2 = 14x + 20y$.

Lemma 1.32 (Euclid's Lemma). *Let $a, b \in \mathbb{Z}$ and p be a prime. Then*

$$p \mid ab \Rightarrow (p \mid a \text{ or } p \mid b).$$

More generally

$$p \mid a_1 \cdots a_k \Rightarrow p \mid a_i \text{ for some } 1 \le i \le k.$$

Proof. For the first part, suppose $p \mid ab$ but p does not divide a. Since p is prime, it is coprime to a, so there exists $\lambda, \mu \in \mathbb{Z}$ such that $1 = \lambda a + \mu p$. Multiply by b to obtain $b = \lambda ab + b\mu p$. We know $p \mid ab$, so p is a factor of the right hand side, so $p \mid b$ as required.

For the second statement, assume that $p \mid a_1 \cdots a_k$. If $k = 1$, then there is nothing to prove. Otherwise, applying the first part with $a = a_1 \cdots a_{k-1}$ and $b = a_k$ allows us to conclude that $p \mid a_1 \cdots a_{k-1}$ or $p \mid a_k$. Repeating this argument $k - 1$ times shows that

$$p \mid a_1 \text{ or } p \mid a_2 \text{ or } \cdots \text{ or } p \mid a_{k-1} \text{ or } p \mid a_k,$$

so $p \mid a_i$ for some $1 \le i \le k$. □

Theorem 1.33 (Fundamental Theorem of Arithmetic). *Every integer that's greater than one can be written uniquely (up to the choice of order) as a product of primes, that is, each $m \in \mathbb{Z}$ satisfying $m \ge 2$ can be written as*

$$m = \prod_{i=1}^{k} p_i = p_1 p_2 \cdots p_k$$

for some $k \ge 1$, where the primes $p_1, \dots, p_k$ are uniquely determined by m.

Proof. The existence of the factorisation was proved in Proposition 1.22. For uniqueness, suppose for contradiction that the decomposition is not always unique. Let m be the minimal criminal, i.e. $m \ge 2$ is the smallest integer admitting distinct factorisations

$$p_1 \cdots p_k = m = q_1 \cdots q_\ell \tag{1.1}$$

with p_i, q_j all primes, and where $k \leq \ell$ are positive integers. Now

$$\begin{aligned} p_1 \mid m \implies p_1 &\mid q_i && \text{for some } i \text{ by Euclid's Lemma,} \\ \implies p_1 &= q_i && \text{as } p_1 \text{ and } q_i \text{ are prime.} \end{aligned}$$

Substitute this into (1.1) to obtain $q_i p_2 \cdots p_k = q_1 \cdots q_\ell$. There are two cases to consider:

1. if $k = 1$, then this is $q_i = q_1 \cdots q_\ell$, and since $q_i \neq 0$, we may divide by q_i to write 1 as a product of primes. This is absurd unless $i = \ell = 1$, so the factorisations of m were the same all along, namely $p_1 = m = q_1$, a contradiction; or

2. otherwise $k \geq 2$. After dividing by $q_i \neq 0$, we have $p_2 \cdots p_k = q_1 \cdots q_{i-1} q_{i+1} \cdots q_\ell$. This gives two factorisations of the number $p_2 \cdots p_k$. However, $p_2 \cdots p_k < m$, which contradicts the minimality of m.

In either case, we conclude that there was one factorisation all along. $\square$

Remark 1.34. Let $m, n \geq 2$ be integers. List the primes in ascending order $p_1 < p_2 < \cdots < p_r$ that arise in the unique factorisations of m and n. Since any given prime may occur more than once, we have

$$m = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r} \quad \text{and} \quad n = p_1^{b_1} p_2^{b_2} \cdots p_r^{b_r}$$

for some $a_i, b_i \in \mathbb{N}_0$. Notice that

1. $m \mid n$ if and only if $a_i \leq b_i$ for $i = 1, \ldots, r$; and

2. the greatest common divisor and least common multiple satisfy

$$\gcd(m,n) = p_1^{\min(a_1,b_1)} \cdots p_r^{\min(a_r,b_r)} \quad \text{and} \quad \operatorname{lcm}(m,n) = p_1^{\max(a_1,b_1)} \cdots p_r^{\max(a_r,b_r)}.$$

Lemma 1.35. *Consider $d, m, n \in \mathbb{N}$. Then:*

(1) $\gcd(m,n) \cdot \operatorname{lcm}(m,n) = m \cdot n$*; and*

(2) $m \mid d$ *and* $n \mid d$ *if and only if* $\operatorname{lcm}(m,n) \mid d$.

Proof. For (1), write the factorisations of m, n as in Remark 1.34. Then since

$$\min(a_j, b_j) + \max(a_j, b_j) = a_j + b_j$$

for each $1 \leq j \leq r$, we have that

$$\begin{aligned} \gcd(m,n) \cdot \operatorname{lcm}(m,n) &= p_1^{\min(a_1,b_1)+\max(a_1,b_1)} \cdots p_r^{\min(a_r,b_r)+\max(a_r,b_r)} \\ &= p_1^{a_1+b_1} \cdots p_r^{a_r+b_r} \\ &= mn. \end{aligned}$$

For (2), we must prove two implication statements, and we choose to begin with the statement that turns out to be easiest to prove. Assume first that $\mathrm{lcm}(m,n) \mid d$. Since $m,n \mid \mathrm{lcm}(m,n)$, we may already conclude that $m \mid d$ and $n \mid d$. For the opposite implication, assume that $m \mid d$ and $n \mid d$. As in Remark 1.34, this time applied to each of d, m and n, we may list primes in ascending order $p_1 < p_2 < \cdots < p_r$ such that

$$m = p_1^{a_1} \cdots p_r^{a_r} \quad \text{and} \quad n = p_1^{b_1} \cdots p_r^{b_r} \quad \text{and} \quad d = p_1^{c_1} \cdots p_r^{c_r}.$$

Since $m \mid d$ and $n \mid d$, we have $a_i, b_i \leq c_i$ for all $1 \leq i \leq r$. Therefore $\max(a_i, b_i) \leq c_i$ for all $1 \leq i \leq r$, from which it follows that the integer

$$\mathrm{lcm}(m,n) = p_1^{\max(a_1,b_1)} \cdots p_r^{\max(a_r,b_r)}$$

divides $p_1^{c_1} \cdots p_r^{c_r} = d$ as required. $\square$

Example 1.36. Let's compute the gcd and lcm of 70 and 600. First, compute the prime factorisations of both numbers, giving

$$70 = 2^1 3^0 5^1 7^1 \quad \text{and} \quad 600 = 2^3 3^1 5^2 7^0.$$

Therefore $\gcd(70, 600) = 2^1 3^0 5^1 7^0 = 10$ and $\mathrm{lcm}(70, 600) = 2^3 3^1 5^2 7^1 = 4200$. As a sanity check, notice that $10 \cdot 4200 = 70 \cdot 600$ which agrees with Lemma 1.35(1).

1.4 Modular arithmetic

Now we introduce and study a third set of numbers; in fact, for each natural number n, we introduce a set of numbers $\mathbb{Z}_n$ that contains only n elements.

Definition 1.37 (Congruence modulo n). For $n \in \mathbb{N}$ and $a, b \in \mathbb{Z}$, we say that a and b are *congruent modulo n* if and only if $n \mid (a-b)$, in which case we write $a \equiv b \pmod{n}$, or simply $[a] = [b]$ if n is understood from the context.

Lemma 1.38. *For $a \in \mathbb{Z}$, there exists r satisfying $0 \leq r < n$ such that $a \equiv r \pmod{n}$, i.e. every integer is congruent modulo n to an integer r in the range $0 \leq r < n$.*

Proof. 'Division by n with remainder' means that for any $a \in \mathbb{Z}$, there exists $q, r \in \mathbb{Z}$ satisfying $a = qn + r$, with $0 \leq r < n$, in which case $a \equiv r \pmod{n}$. $\square$

Definition 1.39 (Congruence classes). Let $n \in \mathbb{N}$. The *congruence classes modulo n* are the n distinct sets

$$\begin{aligned}
[0] &= \{\cdots, -2n, -n, 0, n, 2n, \cdots\} \\
[1] &= \{\cdots, -2n+1, -n+1, 1, n+1, 2n+1, \cdots\} \\
&\vdots \\
[n-1] &= \{\cdots, -2n-1, -n-1, -1, n-1, 2n-1, \cdots\}
\end{aligned}$$

into which $\mathbb{Z}$ is 'partitioned' by the 'congruence modulo n' relation. We write

$$\mathbb{Z}_n = \{[a] \mid a \in \mathbb{Z}\} = \big\{[0], [1], \ldots, [n-1]\big\}$$

for the set of all congruence classes modulo n.

We now show that $\mathbb{Z}_n$ becomes a system of numbers in which we can do arithmetic.

Lemma 1.40. *There are well-defined sum and product operations on $\mathbb{Z}_n$ given by*

$$[a] + [b] = [a + b] \quad \textit{and} \quad [a] \cdot [b] = [a \cdot b].$$

Proof. We need to show these definitions are independent of the representatives chosen, that is, if $a_1 \equiv a_2 \pmod{n}$ and $b_1 \equiv b_2 \pmod{n}$, then we must show that we get the same sum and product irrespective of whether we pick a_2 rather than a_1, and b_2 rather than b_1. Explicitly, we must show that

$$a_1 + b_1 \equiv a_2 + b_2 \pmod{n} \quad \text{and} \quad a_1 b_1 \equiv a_2 b_2 \pmod{n}.$$

Now, if $a_2 = a_1 + qn$ and $b_2 = b_1 + pn$, then

$$\begin{aligned} a_2 + b_2 &= a_1 + b_1 + (q + p)n \\ a_2 b_2 &= (a_1 + qn)(b_1 + pn) = a_1 b_1 + (qb_1 + a_1 p + qpn)n. \end{aligned}$$

The first equation tells us that $[a_2 + b_2] = [a_1 + b_1]$, while the second equation tells us that $[a_2 b_2] = [a_1 b_1]$ as required. □

Remark 1.41. Just as $\mathbb{C}$ inherits nice properties from $\mathbb{R}$, the basic rules of arithmetic[5] are passed down from $\mathbb{Z}$ to $\mathbb{Z}_n$, giving $\mathbb{Z}_n$ many nice properties. For example, sum and product are associative and commutative. Also

$$[x] + [0] = [x + 0] = [x], \; [x] \cdot [1] = [x \cdot 1] = [x] \text{ and } [x] + [-x] = [x + (-x)] = [0]$$

are all special cases of the equations in Lemma 1.40.

Proposition 1.42. *Let $n \in \mathbb{N}$ and $m \in \mathbb{Z}$. Then $[m] \in \mathbb{Z}_n$ has a multiplicative inverse if and only if m and n are coprime.*

Proof. Note that $[m]$ has a multiplicative inverse if and only if there exists $\lambda \in \mathbb{Z}$ such that $[\lambda][m] = [1] \in \mathbb{Z}_n$, that is, if and only if $\lambda m \equiv 1 \pmod{n}$, which, in turn, holds if and only if there exists $\lambda, \mu \in \mathbb{Z}$ such that $\lambda m = 1 + (-\mu)n$, which holds if and only if there is a Bézout identity $\lambda m + \mu n = 1$, which holds if and only if $\gcd(m, n) = 1$. □

Corollary 1.43. *If p is prime, then each of $[1], [2], \ldots, [p-1]$ have a multiplicative inverse in $\mathbb{Z}_p$. That is, every nonzero congruence class in $\mathbb{Z}_p$ has a multiplicative inverse.*

Proof. This follows from Proposition 1.42 since $1, 2, \ldots, p - 1$ are all coprime to p. □

This corollary tells us that $\mathbb{Z}_n$ behaves very much like the more familiar systems of numbers $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ when n is prime; these number systems are all *fields* (see Section 3.1). However, Proposition 1.42 also tells us that $\mathbb{Z}_n$ behaves quite differently when n is not prime; it is a *ring* that is not a *field* (see Section 3.1).

We say much more about inverse maps in Section 2.3, but for now let's record a useful observation:

[5] An object with sum and product operations that satisfies certain natural properties (like 'addition and multiplication are both associative', etc) is called a *ring*; see Section 3.1.

Corollary 1.44 (**Multiplication by a coprime element is invertible**). *If $m, n \in \mathbb{Z}$ are coprime, then the map $M_m\colon \mathbb{Z}_n \to \mathbb{Z}_n$ defined by sending an element $[a]$ to*

$$M_m\big([a]\big) = [a \cdot m]$$

has an inverse map.

Proof. Since m is coprime to n, Proposition 1.42 shows that $[m]$ has a multiplicative inverse in $\mathbb{Z}_n$, say $[\lambda]$. We claim that the map $M_\lambda\colon \mathbb{Z}_n \to \mathbb{Z}_n$ defined as multiplication by $[\lambda]$ is the inverse map to M_m. Indeed, for any $a, b \in \mathbb{Z}$, we have that

$$\begin{aligned}(M_\lambda \circ M_m)[a] &= M_\lambda\Big(M_m\big([a]\big)\Big) = [\lambda] \cdot [m \cdot a] = [1] \cdot [a] = [a] \\ (M_m \circ M_\lambda)[b] &= M_m\Big(M_\lambda\big([b]\big)\Big) = [m] \cdot [\lambda \cdot b] = [1] \cdot [b] = [b].\end{aligned}$$

This is precisely what it means for M_λ to be the inverse to M_m (compare Definition 2.36), so we're done. □

We now present a beautiful application of Corollary 1.44.

Theorem 1.45 (**Fermat's Little Theorem**). *If p is prime, then*

$$a^p \equiv a \pmod{p} \quad \text{for all } a \in \mathbb{Z}.$$

Furthermore if p does not divide a, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Let $a \in \mathbb{Z}$. If p does divide a, then $[a] = [0] \in \mathbb{Z}_p$, so $[a]^p = [0]^p = [0] = [a]$, so the first statement holds.

Otherwise, p does not divide a, i.e. $[a] \neq [0]$. Corollary 1.44 shows that multiplication by $[a]$ in $\mathbb{Z}_p$ is invertible. This implies that the set of all numbers $[0], [1], [2], \ldots [p-1]$ in $\mathbb{Z}_p$ is equal to the set obtained by multiplying each of these numbers by $[a]$, namely, the set of numbers $[a][0], [a][1], [a][2], \ldots [a][p-1]$ in $\mathbb{Z}_p$. If we remove $[0] = [a \cdot 0] = [a][0]$ from both sets, then the product in $\mathbb{Z}_p$ of the numbers in either set is the same, i.e.

$$[1] \cdot [2] \cdots [p-1] = [a][1] \cdot [a][2] \cdots [a][p-1] = \big([1] \cdots [p-1]\big)[a]^{p-1}. \tag{1.2}$$

The product $[1] \cdots [p-1]$ is nonzero in $\mathbb{Z}_p$ because $1 \cdot 2 \cdots (p-1)$ in $\mathbb{Z}$ is not divisible by the prime p, so $[1] \cdots [p-1]$ has a multiplicative inverse in $\mathbb{Z}_p$, say $[b]$. Multiply (1.2) on both sides by $[b]$ to obtain $[1] = [a]^{p-1}$, i.e. $a^{p-1} \equiv 1 \pmod{p}$. In particular it follows that $a^p = a \cdot a^{p-1} \equiv a \cdot 1 = a \pmod{p}$. □

Example 1.46. Let's simplify $[3]^{38}$ in $\mathbb{Z}_{13}$. Here we are working with the prime $p = 13$, so the exponent in Fermat's Little Theorem is $p - 1 = 12$, giving $[3]^{12} = [1]$. Therefore

$$[3]^{38} = [3]^{12 \cdot 3 + 2} = [1]^3 [3]^2 = [9].$$

We now turn our attention to solving linear equations in $\mathbb{Z}_n$.

Definition 1.47 (Congruence equation). Let $n \in \mathbb{N}$ and $a, b \in \mathbb{Z}$. Any statement of the form '$ax \equiv b \pmod{n}$' is a *congruence equation.* Solving any such equation means finding all $x \in \mathbb{Z}$ that satisfy the relation, or equivalently, solving $[a][x] = [b]$ in $\mathbb{Z}_n$.

First, let's investigate how to solve $ax \equiv b \pmod{n}$ when a and n are coprime. We know from Proposition 1.42 that $[a]$ has a multiplicative inverse, say $[c]$, in $\mathbb{Z}_n$. Therefore

$$[cb] = [c][b] = [c][a][x] = [x],$$

and the solution is simply $[x] = [cb]$, or equivalently, $x \equiv cb \pmod{n}$. In practice, we compute the multiplicative inverse using Euclid's algorithm, working backwards with Euclid's equation for finding the $\gcd(a, n)$.

Example 1.48. Let's solve the congruence equation

$$3x \equiv 7 \pmod{13}.$$

To find a multiplicative inverse of $[3]$ in $\mathbb{Z}_{13}$, we first compute the equations that arise in the course of Euclid's algorithm for finding $\gcd(3, 13) = 1$, namely

$$\begin{aligned} \underline{13} &= 4 \cdot \underline{3} + \underline{1} \\ \underline{3} &= 3 \cdot \underline{1} + 0. \end{aligned}$$

We then work with these backwards to get a Bézout identity, in this case $\underline{1} = \underline{13} + (-4) \cdot \underline{3}$. This means that $(-4) \cdot 3 \equiv 1 \pmod{13}$ and therefore $[-4] = [9]$ is the multiplicative inverse of $[3]$. Now we proceed as above: multiply our original equation $[3][x] = [7]$ in $\mathbb{Z}_{13}$ only both sides by $[9]$ to get (using $[9][3] = [1]$)

$$[x] = [9] \cdot [7] = [63] = [11]$$

in $\mathbb{Z}_{13}$. Thus the solution is $x \equiv 11 \pmod{13}$.

We now show that solving a congruence equation in the special case when a and n are coprime is simply the second step in solving the general case.

Proposition 1.49 (Solving congruences). *For $n \in \mathbb{N}$ and $a \in \mathbb{Z}$, let $g = \gcd(a, n)$. Then*

$$ax \equiv b \pmod{n} \quad \text{has a solution if and only if } g \mid b,$$

in which case, the congruence reduces to $\frac{a}{g}x \equiv \frac{b}{g} \pmod{\frac{n}{g}}$, where $\frac{a}{g}$ and $\frac{n}{g}$ are coprime.

Proof. The congruence $ax \equiv b \pmod{n}$ is equivalent to $ax - b = kn$ for some $k \in \mathbb{Z}$, that is, $b = ax - kn$. As g is a common divisor of a and n, we see that g must divide b if there is to be a solution. To prove the final statement, suppose that $g \mid b$. The equation is equivalent to $\frac{a}{g}x - \frac{b}{g} = k\frac{n}{g}$ for some $k \in \mathbb{Z}$, so we must solve $\frac{a}{g}x \equiv \frac{b}{g} \pmod{\frac{n}{g}}$. □

Example 1.50. The congruence equation $3x \equiv 5 \pmod 6$ has no solution because $\gcd(3,6) = 3$ does not divide 5.

Example 1.51. Now consider $3x \equiv 9 \pmod 6$. In this case, $\gcd(3,6) = 3$ does divide 9. Dividing through by 3 shows that we must solve $x \equiv 3 \pmod 2$, which in turn is equivalent to $x \equiv 1 \pmod 2$. Thus, the solution is the set of all odd integers.

We now consider how to solve simultaneous congruence equations.

Theorem 1.52 (**Chinese Remainder Theorem**). *Let $m, n \in \mathbb{N}$ be coprime. For any $a, b \in \mathbb{Z}$, the congruence equations*

$$\begin{aligned} x &\equiv a \pmod m \\ x &\equiv b \pmod n \end{aligned}$$

have a unique solution mod mn.

Proof. The solutions in $\mathbb{Z}$ to $x \equiv a \pmod m$ are precisely $x = a + mc$ for $c \in \mathbb{Z}$. Hence to also have $x \equiv b \pmod n$ we need

$$mc \equiv b - a \pmod n. \tag{1.3}$$

Since m, n are coprime, Proposition 1.42 shows that $[m]$ has an inverse $[r]$ in $\mathbb{Z}_n$, where $0 \leq r < n$. Now multiply (1.3) by $[r]$ to obtain $[c] = [r(b-a)]$ in $\mathbb{Z}_n$, so $c = r(b-a) + dn$ for some unknown $d \in \mathbb{Z}$ over which we have no control. Substitute c into $x = a + mc$ to obtain $x = a + mr(b-a) + dmn$. Notice that a, b, m, r are all determined uniquely by the initial data a, b, m, n, so the solution $x \equiv a + mr(b-a) \pmod{mn}$ is unique. □

Example 1.53. To solve the simultaneous congruence equations

$$x \equiv 4 \pmod 5 \quad \text{and} \quad x \equiv 3 \pmod 7,$$

notice that x is a solution to the first equation if and only if

$$x = 4 + 5c$$

for some $c \in \mathbb{Z}$. Substitute in $4 + 5c$ for x in the second congruence to get $5c + 4 \equiv 3 \pmod 7$ or equivalently

$$5c \equiv -1 \pmod 7 \equiv 6 \pmod 7.$$

As 5 and 7 are coprime we know that $[5]$ has a multiplicative inverse in $\mathbb{Z}_7$. We could find it using Euclid's algorithm, but in this case one observes that $3 \cdot 5 = 15 \equiv 1 \pmod 7$ and thus the inverse of $[5]$ is $[3]$ in $\mathbb{Z}_7$. Multiplying $[5][c] = [6]$ by $[3]$ gives

$$[c] = [18] = [4]$$

in $\mathbb{Z}_7$. Thus c is a solution if and only if $c = 4 + 7d$ for some $d \in \mathbb{Z}$, in which case x is a solution to both congruence equations if and only if

$$x = 4 + 5c = 4 + 5(4 + 7d) = 24 + 35d$$

for some $d \in \mathbb{Z}$. Therefore, the solution is $x \equiv 24 \pmod{35}$.

Definition 1.54 (Group of units and Euler's ϕ-function). Let $n \in \mathbb{N}$. Define

- the *group of units* mod n to be

$$\mathbb{Z}_n^* = \{a \in \mathbb{Z}_n \mid a \text{ is invertible}\};$$

- Euler's *'phi'-function* to be the function $\phi\colon \mathbb{N} \to \mathbb{N}$ defined by sending n to the number of integers $a \in \{1, \ldots, n-1\}$ for which $\gcd(a, n) = 1$.

Remark 1.55. By Proposition 1.42, we may equally well write

$$\mathbb{Z}_n^* = \big\{[a] \mid 1 \leq a \leq n-1 \text{ and } \gcd(a,n) = 1\big\}.$$

Thus, Euler's ϕ-function counts the number of elements in the group of units mod n, or equivalently, the number of primitive nth roots of unity (compare Assignment 1, Q6).

Health Warning 1.56. Please make sure you are in a quiet room with no distractions before reading this warning because the case $n = 1$ is a little strange and requires care. Observe that $\mathbb{Z}_1 = \{[0]\}$, but here $[0] = [1]$. We also know that $[1]$ has a multiplicative inverse, because, erm, $[1]^2 = [1^2] = [1]$. This means that $[0]$ has a multiplicative inverse. This shows that $\mathbb{Z}_1^* = \mathbb{Z}_1$, and hence $\mathbb{Z}_1^*$ contains $1 = \phi(1)$ element. Now lie down.

Lemma 1.57 (The group of units is a group). *If $[a], [b] \in \mathbb{Z}_n^*$, then $[ab] \in \mathbb{Z}_n^*$.*

Proof. If $[c]$ and $[d]$ are the multiplicative inverses to $[a]$ and $[b]$ respectively in $\mathbb{Z}_n$, then commutativity of multiplication in $\mathbb{Z}$ shows that $[ab][cd] = [ac][bd] = [1] \cdot [1] = [1]$. □

This lemma allows us to prove the following generalisation of Fermat's Little Theorem.

Theorem 1.58 (Euler). *If $n \in \mathbb{N}$ and $a \in \mathbb{Z}$ is coprime to n, then*

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Proof. List the invertible elements of $\mathbb{Z}_n$ as $[m_1], \ldots, [m_d]$ for $d = \phi(n)$. Since a is coprime to n, Corollary 1.44 shows that the map $M_a\colon \mathbb{Z}_n \to \mathbb{Z}_n$ defined to be multiplication by $[a]$ has an inverse. Lemma 1.57 tells us that if $[m] \in \mathbb{Z}_n$ is invertible, then so is $[a \cdot m]$; in other words, if $[m] \in \mathbb{Z}_n^*$, then $[a \cdot m] \in \mathbb{Z}_n^*$. Thus, the map

$$M_a\colon \mathbb{Z}_n^* \longrightarrow \mathbb{Z}_n^*.$$

satisfying $M_a([m]) = [a \cdot m]$ also has an inverse. As in the proof of Theorem 1.45, the elements $[m_1], \ldots, [m_d]$ in $\mathbb{Z}_n^*$ are the same (though perhaps in a different order) as

$$[a][m_1], \ldots, [a][m_d],$$

so the product of all d elements in $\mathbb{Z}_n^*$ can be computed in either of two ways, namely

$$[m_1] \cdots [m_d] = \big([a][m_1]\big) \cdots \big([a][m_d]\big) = [a]^d [m_1] \cdots [m_d].$$

Multiplying on both sides by the inverse of $[m_1 \cdots m_d] = [m_1] \cdots [m_d] \in \mathbb{Z}_n^*$ gives $[1] = [a]^d$, so $a^d \equiv 1 \pmod{n}$, that is, $a^{\phi(n)} \equiv 1 \pmod{n}$. □

2 Sets and Functions

Sets and functions (or maps) between them provide the language in which much of modern mathematics is expressed, particularly modern algebra. I'll go quickly through the basics, but you may find some of this content useful as a reference.

2.1 Naive set theory

Definition 2.1 (Sets). A *set* is a collection of things, called its *elements*. We write $x \in A$ to mean that x is an element of the set A, and $x \notin A$ to mean that it isn't.

This isn't really a definition at all, because we haven't defined what 'things' are. The important point to take on board is that a set is characterised by the elements that it contains, so two sets containing the same elements are the same set. In other words

$$A = B \text{ if and only if } \; x \in A \Leftrightarrow x \in B.$$

To put it another way, once you have specified a collection of elements, then you have defined a set.

Remark 2.2.

1. A set is itself a 'thing', so can be an element of another set.
2. A set is normally specified by writing all of its elements between curly brackets.
3. A set need not contain any elements at all. As we show below, there is a unique set containing no elements, denoted $\varnothing$ rather than $\{\ \}$.
4. Repetitions and reordering don't change a set, e.g. $\{1, 2, 2\} = \{1, 2\} = \{2, 1\}$.

Examples 2.3. Define $A = \{1, 2, 5\}$ and $B = \{\{1, 2\}, \{3, 4, 5\}\}$.

1. The set A contains three elements, all of which are numbers. Note that $5 \in A$, but $4 \notin A$.
2. The set B contains two elements, namely, the sets $\{1, 2\}$ and $\{3, 4, 5\}$. Note that $B \neq \{1, 2, 3, 4, 5\}$, where the latter is a set with five elements, all of which are numbers.
3. Note that $\{2\} \neq 2$ and $\{\{2\}\} \neq \{2\}$.

Health Warning 2.4. We've seen that the elements in a set may themselves be sets. However, the set of all sets cannot exist[6], so one must be careful in establishing set theory. This explains why our Definition 2.1 is deliberately vague.

Definition 2.5 (Subsets). Let A and B be sets. We say that

[6] The curious reader might want to investigate *Russell's paradox* at this point. This is non-examinable, but the publication of this observation by Bertrand Russell in 1901 proved to be an important step in the process of establishing firm logical foundations for Set Theory.

- A is a *subset* of B, denoted $A \subseteq B$, if every element of A also lies in B.

- A is *empty* if A has no elements, i.e. $a \in A$ is never true.

Remark 2.6. It follows that $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. This is often used in practice to show that two sets are equal.

Example 2.7. Compare the sets of numbers that we introduced at the start of this unit. We have inclusions

$$\mathbb{N} \subseteq \mathbb{N}_0 \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C},$$

where we regard $\mathbb{Z} \subseteq \mathbb{Q}$ via the map sending $n \mapsto \frac{n}{1}$, we regard $\mathbb{Q} \subseteq \mathbb{R}$ via the map sending $\frac{m}{n}$ to the decimal expansion of $\frac{m}{n}$ (see Analysis 1A and 'Sequences and functions'), and we regard $\mathbb{R} \subseteq \mathbb{C}$ via the map sending $x \mapsto x + i \cdot 0$.

Lemma 2.8 (There can be only one!). *If A is empty, then $A \subseteq B$ for any set B. In particular, there is precisely one empty set, denoted $\varnothing$.*

Proof. We want to show that every element of A is an element of B. The only way this would not be true is if we could find an element in A that is not in B. This is however impossible as A has no elements, so the first statement holds. For the second, suppose both A and B are empty. Then $A \subseteq B$ and $B \subseteq A$ by the first statement, so $A = B$. $\square$

Definition 2.9 (Power set). For any set A, the *power set*, denoted $\mathcal{P}(A)$, is the set of all subsets of A. Thus $B \in \mathcal{P}(A) \iff B \subseteq A$.

Remark 2.10. For any set A, we have $\varnothing \subseteq A$ and $A \subseteq A$, so $\varnothing, A \in \mathcal{P}(A)$.

Example 2.11. If $A = \{1, 2\}$, then $\mathcal{P}(A) = \{\varnothing, \{1\}, \{2\}, \{1, 2\}\}$.

We can describe a subset of a set A by specifying a meaningful property that characterises its elements, i.e.

$$B = \{a \in A \mid \mathbb{P}(a)\},$$

where $\mathbb{P}(a)$ denotes some statement about "a" that either holds or doesn't hold.

Example 2.12. The set of even integers can be described as

$$\{x \in \mathbb{Z} \mid x \text{ is divisible by } 2\}.$$

It is important here to give the set of things to which the property should apply, in this case, the set of integers. Simply writing $\{x \mid x \text{ is divisible by } 2\}$, is not meaningful unless we specify immediately where they belong (as in the next definition!).

Definition 2.13 (Unions, intersections and differences). Given sets A and B we can form their

- *union* $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$;

- *intersection* $A \cap B = \{x \mid x \in A \text{ and } x \in B\} = \{x \in A \mid x \in B\} = \{x \in B \mid x \in A\}$;

- *difference* $A \setminus B = \{x \in A \mid x \notin B\}$. In the special case where $B \subseteq A$, the difference $A \setminus B$ is called the *complement of B in A*.

More generally, given an indexed collection of sets $\{A_j \mid j \in J\}$, where J is some non-empty index set, e.g. $J = \{1, \ldots, n\}$, we can form their

- *union*, defined to be $\displaystyle\bigcup_{j \in J} A_j = \{x \mid x \in A_j \text{ for some } j \in J\}$

- *intersection*, defined to be $\displaystyle\bigcap_{j \in J} A_j = \{x \mid x \in A_j \text{ for all } j \in J\}$

Example 2.14. Suppose that $A_1 = \{5, 6\}$, $A_2 = \{6, 7\}$ and $A_3 = \{6, 8, 9\}$, and define $J = \{1, 2, 3\}$. Then

$$A_1 \cap A_2 = \{6\}; \qquad \bigcup_{j \in J} A_j = \bigcup_{j=1}^{3} A_j = \{5, 6, 7, 8, 9\}; \qquad \bigcap_{j \in J} A_j = \bigcap_{j=1}^{3} A_j = \{6\}$$

Remark 2.15. Each part of Definition 2.13 has an equivalent logical form, such as

$$x \in A \cup B \iff x \in A \text{ or } x \in B,$$

which can be used to prove basic identities for set operations, such as commutativity $A \cup B = B \cup A$ and associativity $(A \cup B) \cup C = A \cup (B \cup C)$. One proves these by showing that the sets have the same elements, e.g.

$$x \in A \cup B \iff x \in A \text{ or } x \in B \iff x \in B \text{ or } x \in A \iff x \in B \cup A.$$

As well as sets, you'll likely be familiar with sets comprising *ordered pairs, triples or n-tuples*, in which each element is itself a list of elements (!). We use round brackets here, so for example

$$\begin{aligned}
(a, b) = (x, y) &\iff a = x \text{ and } b = y \\
(a, b, c) = (x, y, z) &\iff a = x,\ b = y \text{ and } c = z \\
(a_1, \ldots, a_n) = (x_1, \ldots, x_n) &\iff a_i = x_i \text{ for } i = 1, \ldots, n
\end{aligned}$$

Crucially, the order and number of entries is important, so $(1, 2) \neq (2, 1) \neq (2, 1, 1)$.

Definition 2.16 (Cartesian product). The *Cartesian product*[7] of two sets A and B is defined to be

$$A \times B = \big\{(a, b) \mid a \in A,\ b \in B\big\}.$$

[7] The term "Cartesian" is in honour of the French mathematician and philosopher Descartes, who observed that you can model the plane by $\mathbb{R}^2$ and three-dimensional space by $\mathbb{R}^3$. You can then model subsets of $\mathbb{R}^2$, etc. using equations and inequalities. For example,

- $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = r^2\}$ is the circle radius r centred at $(0, 0)$,
- $\{(x, y) \in \mathbb{R}^2 \mid x \geq 0,\ y \geq 0\}$ is the positive (or more accurately, the non-negative) quadrant.

More generally, the *Cartesian product* of sets $A_1, \dots, A_n$ for some $n \geq 1$ is

$$A_1 \times \cdots \times A_n = \big\{(a_1, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, \dots, n\big\}.$$

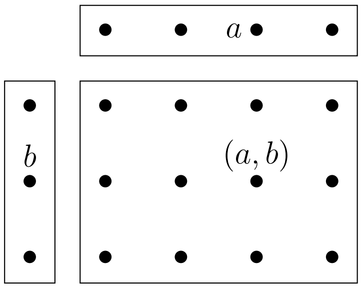
The *Cartesian powers* of a set A are defined to be $A^2 = A \times A$, $A^3 = A \times A \times A$, and more generally, for any $n \geq 1$, we have

$$A^n = \big\{(a_1, \dots, a_n) \mid a_i \in A \text{ for } i = 1, \dots, n\big\}.$$

Example 2.17. For $A = \{1, 3\}$ and $B = \{4, 5\}$, we have that

$$A \times B = \big\{(1,4), (1,5), (3,4), (3,5)\big\} \quad \text{and} \quad A^2 = \big\{(1,1), (1,3), (3,1), (3,3)\big\}.$$

For another example, if A has precisely 4 elements and B has 3, then $A \times B$ has 12:



2.2 Functions

Definition 2.18 (Function). Given sets A and B, a *function* (also called a *map*)

$$f \colon A \longrightarrow B$$

is a rule that assigns to every element $a \in A$, a unique element $f(a) \in B$. We call A the *domain* of f and B the *codomain* of f.

Remarks 2.19.

1. We sometimes write simply 'f' instead of '$f \colon A \to B$' if it is clear from the context what A and B are. However, if A and B are not clear from the context, then writing simply 'f', or even '$f(x) = x^2$', does not specify a function.

2. Two functions f, g are equal if and only if they have the same domain A, the same codomain B *and* they satisfy $f(a) = g(a)$ for all $a \in A$. Thus for example, the rule $f(x) = x^2$ determines different functions $f \colon \mathbb{Q} \to \mathbb{Q}$, $f \colon \mathbb{Z} \to \mathbb{Q}$, $f \colon \mathbb{Z} \to \mathbb{Z}$.

Functions are determined by their graph.

Definition 2.20 (Graph of a function). Given a function $f \colon A \to B$, the *graph* of f is the subset of $A \times B$ given by

$$\mathrm{Graph}(f) = \{(a, b) \in A \times B \mid b = f(a)\}.$$

Remark 2.21. The sets A, B and a subset G of $A \times B$, determine a function $f : A \to B$ if and only if G satisfies the 'graph property': *for each* $a \in A$*, there is a unique* $b \in B$ *such that* $(a, b) \in G$. When this is the case, this unique b is then our $f(a)$.

Examples 2.22.

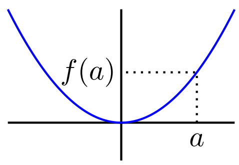
1. If $A = \{1\}$ and $B = \{2\}$, then there is only one possible function $f\colon A \to B$. It must map 1 to 2. Here $\text{Graph}(f) = \{(1, 2)\}$.

2. If $A = \{1\}$ and $B = \{1, 2\}$, then there are two possible functions with domain A and codomain B. These are $f\colon A \to B$ that maps 1 to 1 and $g\colon A \to B$ that maps 1 to 2. Here $\text{Graph}(f) = \{(1, 1)\}$ and $\text{Graph}(g) = \{(1, 2)\}$.

3. If $A = \{1\}$ and $B = \varnothing$ then there is no function with domain A and codomain B because we can't map 1 to an element in B.

4. If $A = \varnothing$ and B is any set, then there is a unique function f with domain A and codomain B. You can think of this as the 'lazy machine' that doesn't need to produce anything because the domain is empty. Notice that for every $a \in A$, there is a unique $f(a) \in B$ (simply because there is no contradiction!). Here $\text{Graph}(f) = \varnothing$.

5. Consider the function $f\colon \mathbb{R} \to \mathbb{R}$ where $f(x) = x^2$, so $\text{Graph}(f) = \{(x, x^2) : x \in \mathbb{R}\}$. In this last example, we can visualise the graph by using the cartesian coordinate



system for $\mathbb{R}^2$, but this is not possible in general.

Definition 2.23 (Identity, composition and inverse). Let A, B, C be sets.

- The *identity map* id_A is the map whose domain and codomain both equal A, and where $\text{id}_A(a) = a$ for all $a \in A$.

- Given functions $f\colon A \to B$ and $g\colon B \to C$, their *composition* $(g \circ f)\colon A \to C$, which we read as '$g$ follows f', is the function given by

$$(g \circ f)(a) = g(f(a)) \quad \text{for } a \in A..$$

  We do not consider $g \circ f$, unless the codomain of f and the domain of g coincide.

- A function $f\colon A \to B$ is said to be *invertible* (or *to have an inverse*) if there exists a function $g\colon B \to A$ such that $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$.

Remark 2.24 (Composition of functions is associative). Given functions $f\colon A \to B$, $g\colon B \to C$ and $h\colon C \to D$, we have

$$h \circ (g \circ f) = (h \circ g) \circ f$$

because both sides are the function $A \to D$ that maps a to $h(g(f(a)))$. We may therefore write this triple composition simply as $h \circ g \circ f$ without ambiguity, and similarly for compositions created from more than three functions. In particular, for the case of a function $f\colon A \to A$ whose domain and codomain agree, we can form $f^n = f \circ \cdots \circ f$, where f is composed with itself n times, for any $n \in \mathbb{N}$. Usually one defines $f^0 = \mathrm{id}_A$.

Functions are also things, so can be elements of sets.

Definition 2.25 (Function spaces). Let A and B be sets. The set of *functions from A to B* is denoted

$$B^A = \{\text{functions } A \to B\}.$$

Remark 2.26. Since each function is determined by its graph (and conversely), we may equally well regard B^A as the set

$$\{G \subseteq A \times B \mid G \text{ has the graph property}\}.$$

of all possible graphs of functions from A to B.

Examples 2.27 (Some standard functions associated to subsets). This seems like a good time to introduce some standard functions, each associated to any subset of a set.

1. For any set A, we can identify $\mathcal{P}(A)$ with the set $\{0,1\}^A$ of functions from A to $\{0,1\}$ by associating to each subset $S \subseteq A$, its *characteristic function*

$$\chi_S\colon A \to \{0,1\}\colon a \mapsto \begin{cases} 1 & \text{if } a \in S \\ 0 & \text{if } a \notin S \end{cases}.$$

   Conversely, we can recover S from χ_S via $S = \{a \in A \mid \chi_S(a) = 1\}$. That is, when working with the power set of A, we may instead think about the set $\{0,1\}^A$ of functions from A to $\{0,1\}$ if that helps.

2. For any subset $S \subseteq A$, there is an *inclusion map* $\mathrm{inc}_S^A\colon S \to A$ that sends $x \in S$ to $x \in A$ (!).

3. Let $f\colon A \to B$ be any map and $S \subseteq A$ a subset. The *restriction of f to S*, denoted $f|_S\colon S \to B$, is defined to be the function $f|_S(x) = f(x)$ for all $x \in S$. In other words, the rule for $f|_S$ is the same as the rule for f, but the domain is smaller.

2.3 Injective, surjective and bijective

Our next goal is to introduce several key notions in the study of functions.

Definition 2.28 (Image of a function). Let $f\colon A \to B$ be a function. The *image of f*, denoted $\mathrm{Im}(f)$, is the subset of B given by

$$\mathrm{Im}(f) = \big\{b \in B \mid b = f(a) \text{ for some } a \in A\big\} = \big\{f(a) \mid a \in A\big\}.$$

Now, the three key definitions.

Definition 2.29 (Injective, surjective and bijective). Let $f\colon A \to B$ be a map. We say that f is:

- *injective* if whenever $a, a' \in A$ satisfy $f(a) = f(a')$, it follows that $a = a'$;
- *surjective* if $\operatorname{Im}(f) = B$; and
- *bijective* precisely when f is both injective and surjective.

Remark 2.30. A map $f\colon A \to B$ is surjective when the equation $y = f(x)$ has a solution for every $y \in B$; it is injective when the solution is unique whenever it exists. Thus f *is bijective if the equation $y = f(x)$ has a unique solution for every $y \in B$.*

Examples 2.31.

1. The map $f\colon \mathbb{R} \to \mathbb{R}\colon x \mapsto x^2$ is not injective because $x = \pm 1$ both satisfy $x^2 = 1$, and it is not surjective because -1 is not of the form x^2 for any $x \in \mathbb{R}$. However if we replace the codomain of f by the set
$$\operatorname{Im}(f) = \{y \in \mathbb{R} \mid y \geq 0\},$$
then the resulting map $g\colon \mathbb{R} \to \operatorname{Im}(f)\colon x \mapsto x^2$ is now surjective. Indeed replacing the codomain of any map by its image makes the resulting map surjective.

2. The map $h\colon \mathbb{R} \to \mathbb{R}\colon x \mapsto 3x + 2$ is bijective because the equation $y = 3x + 2$ has a unique solution for every $y \in \mathbb{R}$, namely $x = (y - 2)/3$.

Remarks 2.32.

1. When stated in logical terms, f is injective iff
$$\forall\, a, a' \in A, \text{ we have } f(a) = f(a') \implies a = a'.$$
Thus, to prove that f is injective, the direct approach is to first write "*Let $a, a' \in A$ satisfy $f(a) = f(a')$*", before going on to show that $a = a'$ after all.

2. A function fails to be injective when the negation of this statement holds, i.e.
$$\exists\, a, a' \in A \text{ such that } f(a) = f(a') \text{ and } a \neq a'.$$

3. Visually, a function $f\colon \mathbb{R} \to \mathbb{R}$ is injective if any horizontal line cuts the graph of f in at most one point. It is bijective if any horizontal line cuts the graph of f in precisely one point. However, that intuition may not be helpful for functions with other (co)domains, so do please get to grips with the definition.

Examples 2.33.

1. Consider $f\colon \{1, 2\} \to \{1, 2, 3\}$ defined by $f(1) = 2, f(2) = 1$. This function is injective as $f(1) \neq f(2)$. However, it is not surjective because 3 is not an output value of f, i.e. $3 \notin \operatorname{Im}(f)$. It follows then that f is not bijective.

2. Consider $g\colon \{1,2,3\} \to \{1,2\}$ defined by $g(1) = g(2) = 1$ and $g(3) = 2$. This function is not injective as $g(1) = g(2)$, but it is surjective as each element in the codomain is an output value of g, i.e. $\operatorname{Im}(g) = \{1,2\}$. Again, g is not bijective.

3. Consider $h\colon \{1,2,3\} \to \{4,5,6\}$ defined by $h(1) = 5, h(2) = 6$ and $h(3) = 4$. This function is injective as distinct elements map to distinct elements, and it is surjective as $\operatorname{Im}(h) = B$. As it is both injective and surjective, h is bijective.

4. Consider $i\colon \{1,2\} \to \{1,2\}$ where $i(1) = i(2) = 1$. This function is not injective as $i(1) = i(2)$, nor is it surjective because $2 \notin \operatorname{Im}(i)$. Thus, i is not bijective.

Theorem 2.34 (**Chinese remainder theorem revisited**). *Let $m, n \in \mathbb{N}$ be coprime. The map $f\colon \mathbb{Z}_{mn} \to \mathbb{Z}_m \times \mathbb{Z}_n$ defined by*

$$f\big(\, x \pmod{mn}\big) = \big(\, x \pmod{m},\ x \pmod{n}\big)$$

is a bijective map.

Proof. This is just a restatement of the existence and uniqueness from Theorem 1.52, compare Remark 2.30. □

Example 2.35. For $m = 2$ and $n = 5$, of course we have $mn = 10$, so the map is a bijection $f\colon \mathbb{Z}_{10} \to \mathbb{Z}_2 \times \mathbb{Z}_5$, but it's really instructive to write down the map explicitly:

$$f([0]) = \big([0],[0]\big); \quad f([1]) = \big([1],[1]\big); \quad f([2]) = \big([0],[2]\big); \quad f([3]) = \big([1],[3]\big);$$
$$f([4]) = \big([0],[4]\big); \quad f([5]) = \big([1],[0]\big); \quad f([6]) = \big([0],[1]\big); \quad f([7]) = \big([1],[2]\big);$$
$$f([8]) = \big([0],[3]\big); \quad f([9]) = \big([1],[4]\big).$$

Notice that all ten elements of $\mathbb{Z}_2 \times \mathbb{Z}_5$ appear exactly once. Pretty, isn't it?!

Definition 2.36 (**Left- and right-inverses**). Let $f\colon A \to B$ be a function. We say that a function $g\colon B \to A$ is:

- a *left-inverse* of f if $g \circ f = \mathrm{id}_A$;
- a *right-inverse* of f if $f \circ g = \mathrm{id}_B$.

Thus, g is an inverse of f (see Definition 2.23) if it is a left- and a right-inverse of f.

Theorem 2.37 (**Existence of left- and right-inverses**). *Consider $f\colon A \to B$ with $A \neq \varnothing$ (and hence $B \neq \varnothing$). Then*

(1) f is injective iff f has a left-inverse $g\colon B \to A$;

(2) f is surjective iff f has a right-inverse $g\colon B \to A$; and

(3) f is bijective iff f has a (two-sided) inverse $g\colon B \to A$.

Proof. For the proof of (1):

($\Longleftarrow$) Let $a, a' \in A$, and suppose that $f(a) = f(a')$. Then since g is a left inverse, we have $a = \mathrm{Id}_A(a) = g(f(a)) = g(f(a')) = \mathrm{Id}_A(a') = a'$ as required.

($\Longrightarrow$) Suppose that f is injective. Fix some $r \in A$, and define $g\colon B \to A$ as follows:

$$g(b) = \begin{cases} a & \text{if } b \in \mathrm{Im}(f), \text{ where } a \in A \text{ is the unique value such that } f(a) = b \\ r & \text{if } b \notin \mathrm{Im(f)} \end{cases}$$

Then for any $a \in A$, the element $b = f(a) \in \mathrm{Im}(f)$ satisfies $g(b) = a$. Thus $g(f(a)) = a$ for all $a \in A$, so g is a left inverse of f.

For the proof of (2):

($\Longleftarrow$) Let $b \in B$. As g is a right inverse, we have $f(g(b)) = b$. This shows that every $b \in B$ is an output value of f, so $\mathrm{Im}(f) = B$ and hence f is surjective.

($\Longrightarrow$) Suppose that f is surjective. Then for each $b \in B$ we can choose some $a \in A$ such that $f(a) = b$. We may therefore define $g\colon B \to A$ by $g(b) = a$. Then $f(g(b)) = f(a) = b$, so g is a right inverse of f.

For the proof of (3):

($\Longleftarrow$) If a two-sided inverse $g\colon B \to A$ exists, then g is both a left and a right inverse. Parts 1. and 2. imply that f is injective and surjective, so f is bijective.

($\Longrightarrow$) If f is bijective, then parts 1. and 2. show that f has a left inverse g and a right inverse h, in which case

$$g = g \circ \mathrm{id}_B = g \circ (f \circ h) = (g \circ f) \circ h = \mathrm{id}_A \circ\ h = h. \tag{2.1}$$

Thus $g = h$ is a two-sided inverse. □

Remark 2.38. The pair of examples to follow illustrate that an injective map $f\colon A \to B$ can have more than one left-inverse, while a surjective map $f\colon A \to B$ can more than one right-inverse.

Example 2.39. Consider $f\colon \{1,2\} \to \{1,2,3\}$ where $f(1) = 2$ and $f(2) = 1$. Define

$$g_1, g_2\colon \{1,2,3\} \to \{1,2\}$$

by $g_1(1) = 2,\ g_1(2) = 1,\ g_1(3) = 1$, and $g_2(1) = 2,\ g_2(2) = 1,\ g_2(3) = 2$. Then

$$(g_i \circ f)(1) = g_i(2) = 1 \quad \text{and} \quad (g_i \circ f)(2) = g_i(1) = 2$$

for $1 \leq i \leq 2$, so $g_i \circ f = \mathrm{id}_{\{1,2\}}$ for $i = 1, 2$, so both g_1 and g_2 are left inverses. This construction of g_1, g_2 is precisely the construction of g from the proof of Theorem 2.37(1).

Example 2.40. For $f\colon \{1,2,3\} \to \{1,2\}$ with $f(1) = f(2) = 1$ and $f(3) = 2$, define

$$g_1, g_2\colon \{1,2\} \to \{1,2,3\}$$

by letting $g_1(1) = 1,\ g_1(2) = 3$, and $g_2(1) = 2,\ g_2(2) = 3$. Then

$$(f \circ g_i)(1) = 1 \quad \text{and} \quad (f \circ g_i)(2) = 2$$

for $1 \leq i \leq 2$, so both g_1 and g_2 are right inverses of f.

Theorem 2.41 (**Inverses are unique**). *Let $f\colon A \to B$ be a bijective map. Then f has a unique (two-sided) inverse that we denote f^{-1}.*

Proof. Suppose g, h are inverses of f. In particular g is a left inverse and h is a right inverse. The argument from (2.1) above shows that $g = h$. □

The third statement in the following result is especially useful.

Proposition 2.42 (**Inverse maps are themselves bijective**). *Let $f\colon A \to B$ be a function such that $A \neq \varnothing$. Then:*

(1) if f is injective, then any left-inverse g is surjective;

(2) if f is surjective, then any right-inverse g is injective.

In particular, if f is bijective, then the two-sided inverse f^{-1} is bijective.

Proof. For (1), we're given that $g \circ f = \mathrm{id}_A$. This means that f is a right-inverse of g, so g is surjective by Theorem 2.37(2). For (2), we're given that $f \circ g = \mathrm{id}_B$, so f is a left-inverse of g and hence g is injective by Theorem 2.37(1). For the final statement, f^{-1} is a left-inverse of f, so it is surjective by (1), and f^{-1} is a right-inverse of f, so it is injective by (2). Thus, f^{-1} is bijective as required. □

Corollary 2.43. *Let $A \neq \varnothing$. There is an injective map $f\colon A \to B$ iff there is a surjective map $g\colon B \to A$.*

Proof. Suppose first that f is injective. If we take g to be a left-inverse to f, then g is surjective by Proposition 2.42(1). Conversely, suppose now that g is surjective. If we take f to be a right-inverse to g, then f is injective by Proposition 2.42(2). □

Remark 2.44. . The assumption that $A \neq \varnothing$ in Proposition 2.42(4) is necessary, because the unique map $f\colon \varnothing \to B$ is injective[8], yet there is no map $g\colon B \to \varnothing$ when $B \neq \varnothing$.

Proposition 2.45. *Let $f\colon A \to B$ and $g\colon B \to C$ be maps. Then:*

(1) if f and g are injective, then so is $g \circ f$,

(2) if f and g are surjective, then so is $g \circ f$,

(3) if f and g are bijective, then so is $g \circ f$.

Proof. The proof for part (3) is an exercise (see Assignment 4). The solution to that exercise establishes parts (1) and (2) in order to establish part (3). □

[8] After all, how could it fail to be injective?!

2.4 Cardinality and counting

Calculating the size of a set involves counting its elements, but this isn't as obvious as it might appear. To make the point, what if John and Jane both count sheep in a field, but list them in different orders? Will they arrive at the same answer? Clearly, we hope so[9].

Only now that we have introduced the notion of a bijection are we in a position to do this properly. We wish to say that a set A has cardinality $n \in \mathbb{N}$ if there is a bijection between A and the set $\{1, \dots, n\}$. However, what if there is also bijection between A and $\{1, \dots, m\}$ for some $m \neq n$? In other words, how do we know that the size of A doesn't depend on the order in which we pick the elements? We address this issue in the non-examinable Appendix A, where we establish the following result:

Proposition 2.46. *Let A be a set. Suppose there are bijective maps $f\colon \{1, \dots, m\} \to A$ and $g\colon \{1, \dots, n\} \to A$. Then $n = m$.*

This result is necessary in order to introduce the following important definition.

Definition 2.47 (Finite set). A set A is *finite* if either $A = \varnothing$, or if there exists $n \in \mathbb{N}$ and a bijective map $f\colon \{1, \dots, n\} \to A$. The *cardinality* of a finite set A is defined to be 0 when $A = \varnothing$, and it is $|A| := n$ when there exists a bijective map $f\colon \{1, \dots, n\} \to A$ (in which case, we may list the elements and write $A = \{a_1, \dots, a_n\}$).

Example 2.48. We have $|\{1, 2, 2\}| = |\{1, 2\}| = 2$, $|\{\{1, 2, 3\}\}| = 1$ and $|\varnothing| = 0$.

The following lemma is intuitively obvious. The proof builds on the properties of the set of natural numbers and our precise definition of cardinality, through induction.

Lemma 2.49. *Let A be a finite set and let $S \subseteq A$. Then S is finite and $|S| \leq |A|$. Furthermore $|S| = |A|$ iff $S = A$.*

Proof. (Non-examinable) We use induction on $n = |A|$. The base case is $n = 0$, in which case $A = \varnothing$ and the only subset is $S = \emptyset$. Hence $|S| = |A| = 0$ and the statement holds.

For the induction step, suppose the statement is true when $|A| = k$. To show it is true for $|A| = k + 1$, write $A = \{a_1, \dots, a_k, a_{k+1}\}$ and let $S \subseteq A$. We consider two cases:

1. suppose that $a_{k+1} \notin S$. Then $S \subseteq A \setminus \{a_{k+1}\} = \{a_1, \dots, a_k\}$. The latter set has cardinality k, so the induction hypothesis implies that $|S| \leq k < k + 1 = |A|$. The second statement also holds because $a_{k+1} \notin S$ and hence $S \neq A$.

2. Otherwise, $a_{k+1} \in S$. Then $S \setminus \{a_{k+1}\} \subseteq A \setminus \{a_{k+1}\} = \{a_1, \dots, a_k\}$. Again by the induction hypothesis we know that $|S \setminus \{a_{k+1}\}| \leq k$ and thus $|S| = |S \setminus \{a_{k+1}\}| + 1 \leq k + 1 = |A|$, so the first statement holds. For the second statement, we know that $S = A$ iff $S \setminus \{a_{k+1}\} = A \setminus \{a_{k+1}\}$. By the inductive hypothesis, the latter holds iff $|S \setminus \{a_{k+1}\}| = |A \setminus \{a_{k+1}\}|$ that holds iff $|S| = |S \setminus \{a_{k+1}\}| + 1 = |A \setminus \{a_{k+1}\}| + 1 = |A|$.

This finishes the induction step and therefore the inductive proof. $\square$

[9] And so do the sheep!

Theorem 2.50. *(1) Let B be a finite set. If $f\colon A \to B$ is injective, then A is also finite and $|A| \leq |B|$. Furthermore, we have $|A| = |B|$ iff f is bijective.*

(2) Let A be a finite set. If $f\colon A \to B$ is surjective, then B is finite and $|A| \geq |B|$. Furthermore, we have that $|B| = |A|$ iff f is bijective.

Proof. For (1), if A is the emptyset, then the statement is obvious (because $\varnothing$ is a finite set, and $0 \leq |B|$ for any finite set B, where of course $|B| = 0$ iff B is also the emptyset). Otherwise, $A \neq \varnothing$. Consider the surjective map $\overline{f}\colon A \to \mathrm{Im}(f), a \mapsto f(a)$ obtained by replacing the codomain of f with its image. This function is bijective since f is injective. The composition of bijective maps is bijective by Proposition 2.45(3), so $|A| = |\mathrm{Im}(f)|$. We know that $|\mathrm{Im}(f)| \leq |B|$ by Lemma 2.49, so $|A| \leq |B|$. Lemma 2.49 also tells us that $|\mathrm{Im}(f)| = |B|$ (that is $|B| = |A|$) iff $\mathrm{Im}(f) = B$, which holds iff f is surjective.

For (2), f has a right inverse $g\colon B \to A$. But then f is a left inverse of g and we know that g is injective. Part (1) gives $|B| \leq |A|$, and moreover, $|B| = |A|$ iff g is bijective. We finish the proof by showing that g is bijective iff f is bijective. Since g is a right-inverse, we know $f \circ g = \mathrm{id}_B$. Now, g is bijective iff g^{-1} exists and is bijective, which holds iff $f \circ g \circ g^{-1} = g^{-1}$ exists and is bijective, which holds iff $f = g^{-1}$ exists and is bijective. □

Corollary 2.51 (**The Pigeonhole Principle**)**.** *Let $m, n \in \mathbb{N}$, with $m < n$. If we put n objects into m boxes, then at least one of the boxes must contain at least two elements.*

Proof. Suppose otherwise. The set of boxes A and the set of objects B have cardinality m and n respectively. By our assumption, the map $f\colon B \to A$ that assigns each object to the appropriate box is an injective map. Since $|B| = n > m = |A|$, Theorem 2.50(1) implies that there does not exist an injective map $f\colon B \to A$, a contradiction. □

Example 2.52. Suppose that we wish to place 65 grains of sand on a chessboard, where each chess square has side of length 3cm. Let's convince ourselves that we must place at least two of the grains of sand no more than $3\sqrt{2}$ cm apart.

Our chessboard comprises 64 squares, each a 3cm $\times$ 3cm square. For each grain of sand, let's pick one of the 64 squares that it touches (any one grain might touch up to four squares, so just choose one of these). This defines a function f from the set of $n = 65$ grains of sand to the set of $m = 64$ squares. As $m < n$, the map f is not injective, so there exists a square containing (parts of) at least two grains. The distance between the two is therefore at most the length of the diagonal, that is $3\sqrt{2}$cm.

Corollary 2.53. *Let A, B be finite sets where $|A| = |B|$. Let $f : A \to B$ be a function. Then f is injective iff f is surjective iff f is bijective.*

Proof. It suffices to prove that f is injective iff f is surjective, because this pair of statements is the definition of bijective. If f is injective, then Theorem 2.50(1) implies that f is surjective, and conversely, if f is surjective then Theorem 2.50(2) implies that f is injective. □

Proposition 2.54 (**Characterising infinite sets**)**.** *A set A is infinite (i.e. not finite) if and only if there is an injective map $f\colon \mathbb{N} \to A$.*

Proof. (Non-examinable) Suppose first that there is an injective map $f\colon \mathbb{N} \to A$. Assume for a contradiction that A is finite, so there is a bijective map $c\colon \{1, \dots, n\} \to A$. For any $m \in \mathbb{N}$, Proposition 2.45(1) implies that the composition

$$c^{-1} \circ f \circ \mathrm{inc}\colon \{1, \dots, m\} \to \{1, \dots, n\}$$

is injective. This contradicts the Pigeonhole Principle when $m > n$, so A must be infinite.

Conversely, if A is infinite, then we can build an injection $f\colon \mathbb{N} \to A$ recursively. That is, we can build injections $f_n\colon \{1, \dots, n\} \to A$, for all $n \in \mathbb{N}$, such that, for all $m \leq n$, the restriction of f_n to $\{1, \dots, m\}$ is f_m. We start with any map $f_1\colon \{1\} \to A$; this is necessarily injective. Given an injective map f_k, extend it to f_{k+1} as follows: since A is not finite, f_k cannot be surjective and hence there exists $a \in A \setminus \mathrm{Image}(f_k)$. Defining $f_{k+1}(k+1) = a$ gives the required map $f_{k+1}\colon \{1, \dots, k+1\} \to A$. Thus, if A is not finite, then this recursive construction works for all $n \in \mathbb{N}$. □

Definition 2.55 (Countable sets). Let A be a set. We say that A is *countably infinite* if there is a bijection $f\colon \mathbb{N} \to A$, and it is *countable* if A is finite or countably infinite.

We will not study this in any detail, but let's record a couple of facts that are good to know (compare Analysis 1A and 'Sequences and functions').

- the sets $\mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}$ and $\mathbb{Q}$ are countably infinite;
- $\mathbb{R}$ is not countably infinite, i.e. it is infinite but not countably so.

2.5 Equivalence relations

We now study a special kind of binary relation on a set. A *binary relation* $\sim$ on a set A is nothing more than a property that any given pair of elements of A either satisfy, or they do not satisfy. For $a, b \in A$, we write $a \sim b$ if the pair (a, b) satisfies the property, and $a \not\sim b$ if the pair does not. A binary relation $\sim$ is determined completely by the set

$$\big\{(a, b) \in A \times A \mid a \sim b\big\}$$

of pairs that satisfy the relation. Therefore, a binary relation on a set A is nothing more complicated than a subset of $A \times A$ (comprising the pairs (a, b) for which $a \sim b$).

Definition 2.56 (Equivalence relation). A binary relation $\sim$ on a non-empty set A is an *equivalence relation* if it satisfies the following three properties:

(R) $\sim$ is *reflexive*: $a \sim a$ for all $a \in A$.

(S) $\sim$ is *symmetric*: $a \sim b \Rightarrow b \sim a$ for all $a, b \in A$.

(T) $\sim$ is *transitive*: $a \sim b$ and $b \sim c \Rightarrow a \sim c$ for all $a, b, c \in A$.

Examples 2.57. 1. On any set, the notion of equality is an equivalence relation (which of course we denote $=$ rather than $\sim$).

2. On the set $\mathbb{R}$, the relation $\leq$ is reflexive and transitive but not symmetric, whereas the relation $<$ is only transitive.

3. On $\mathcal{P}(A) \setminus \{\varnothing\}$ define $\sim$ to be 'overlaps', that is, $X \sim Y$ iff $X \cap Y \neq \varnothing$. This is reflexive and symmetric, but not transitive.

4. Let $f\colon A \to B$ be any function. Define $\sim$ on A by $a \sim a'$ iff $f(a) = f(a')$. Then:

   (R) $\sim$ is reflexive: for $a \in A$, we have $f(a) = f(a)$, so $a \sim a$;

   (S) $\sim$ is symmetric: for $a, b \in A$, if $f(a) = f(b)$, then $f(b) = f(a)$; and

   (T) $\sim$ is transitive: for $a, b, c \in A$, if $f(a) = f(b)$ and $f(b) = f(c)$, then $f(a) = f(c)$.

Definition 2.58 (**Equivalence classes**)**.** Let $\sim$ be an equivalence relation on A. For each $a \in A$, the *equivalence class* of a is defined to be $[a] = \{x \in A \mid x \sim a\}$. Let

$$A/{\sim} = \{[a] \mid a \in A\}$$

denote the *set of equivalence classes.*

Example 2.59. On $\mathbb{R}$, let $x \sim y$ iff $x^2 = y^2$. This equivalence relation gives $[a] = \{a, -a\}$.

Lemma 2.60. *Let $\sim$ be an equivalence relation on a set A. Then $a \sim b \iff [a] = [b]$.*

Proof. Suppose first that $[a] = [b]$. We have $a \in [a]$ since $a \sim a$ (R), so $a \in [b]$ and hence $a \sim b$. For the converse, suppose that $a \sim b$. We must show that $x \in [a] \iff x \in [b]$, i.e. that $x \sim a \iff x \sim b$. For this, notice that if $a \sim b$ and $x \sim a$, then $x \sim b$ by (T). On the other hand, if $a \sim b$ and $x \sim b$, then $b \sim a$ by (S) and so $x \sim a$ by (T). $\square$

Definition 2.61 (**Partition of a set**)**.** A *partition* of a set A is a set of pairwise disjoint non-empty subsets of A whose union is A, i.e. a set $\{A_i\}_{i\in I}$ of non-empty subsets $A_i \subseteq A$ such that $A = \bigcup_{i\in I} A_i$ and $A_i \cap A_j = \varnothing$ for all $i, j \in I$ with $i \neq j$.

Example 2.62. The set $A = \{1, 2, 3\}$ admits several partitions, namely

$$\big\{\{1,2,3\}\big\}, \big\{\{1\},\{2,3\}\big\}, \big\{\{1,3\},\{2\}\big\}, \big\{\{1,2\},\{3\}\big\}, \big\{\{1\},\{2\},\{3\}\big\}.$$

Theorem 2.63 (**Equivalence relation '=' partition**)**.** *Let A be a non-empty set.*

(1) if $\sim$ is an equivalence relation on A, then the set $A/{\sim} = \{[a] \mid a \in A\}$ of equivalence classes forms a partition of A; and

(2) if $\{A_i\}_{i\in I}$ is a partition of A, then the binary relation $\sim$ on A defined by

$$a \sim b \iff \exists\, i \in I \text{ such that } a, b \in A_i \tag{2.2}$$

is an equivalence relation.

These two constructions are mutually inverse, i.e. they provide a natural bijection (and its inverse) between the set of equivalence relations on A and the set of partitions of A.

Proof. For (1), we have $a \in [a]$ for all $a \in A$ by (R), so each equivalence class is nonempty. Also, A is the union of its own elements (!), so it is certainly the union of its equivalence classes $[a]$ over all $a \in A\}$. To see that distinct equivalence classes are disjoint, consider $a, b \in A$ with $[a] \neq [b]$. Suppose for a contradiction that $[a] \cap [b] \neq \varnothing$, say $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. It follows that $a \sim c$ by (S), and hence $a \sim b$ by (T). Lemma 2.60 shows that $[a] = [b]$, a contradiction.

For (2), consider the relation $\sim$ from (2.2). Each $a \in A$ lies in some A_i, so $a \sim a$ for all $a \in A$ and hence $\sim$ is reflexive. Next, if $a \sim b$, then there exists $i \in I$ such that $a, b \in A_i$, but then $b, a \in A_i$ (!) and hence $b \sim a$. Finally, if $a \sim b$ and $b \sim c$, then there exists $i, j \in I$ such that $a, b \in A_i$ and $b, c \in A_j$. However, $A_i \cap A_j = \varnothing$ if $i \neq j$ and yet $b \in A_i \cap A_j$, so in fact $i = j$. Therefore, $a, b, c \in A_i$, so $a \sim c$ as required. $\square$

Example 2.64. For $n \in \mathbb{N}$ and $a, b \in \mathbb{Z}$, define a relation on the set $\mathbb{Z}$ by

$$a \sim b \iff n \mid b - a;$$

that is, $a \sim b$ if and only if a and b are congruent modulo n (compare Definition 1.37). This an an equivalence relation, normally written $a \equiv b \pmod{n}$ as in Assignment 5, and the equivalence classes are the n distinct sets

$$\begin{array}{rcl} [0] & = & \{\cdots, -2n, -n, 0, n, 2n, \cdots\} \\ [1] & = & \{\cdots, -2n+1, -n+1, 1, n+1, 2n+1, \cdots\} \\ & \vdots & \\ [n-1] & = & \{\cdots, -2n-1, -n-1, -1, n-1, 2n-1, \cdots\}, \end{array}$$

and indeed, we noted in Definition 1.39 that $\mathbb{Z}$ *is 'partitioned' by the 'congruence modulo n' relation* into these congruence classes. In this case, the set $\mathbb{Z}/\!\sim$ of equivalence classes (that is, the corresponding partition of $\mathbb{Z}$) is traditionally denoted

$$\mathbb{Z}_n = \{[m] \mid m \in \mathbb{Z}\} = \big\{[0], [1], \ldots, [n-1]\big\}.$$

2.6 Permutations

Definition 2.65 (The set of permutations on n letters). For $n \in \mathbb{N}$, a bijective map $\sigma \colon \{1, \ldots, n\} \to \{1, \ldots, n\}$ is called a *permutation* on n letters[10]. Let S_n denote the set of all permutations on n letters.

Remark 2.66. One traditional way to write down a permutation is

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix}.$$

However, there's a more efficient way that we'll first illustrate in an example.

[10]Yes, I know, I know, these are numbers rather than letters. Don't shoot the messenger!

Example 2.67. The permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 2 & 1 & 6 \end{pmatrix}$$

cyclically permutes the elements $\{1, \ldots, 6\}$ in three 'orbits', namely

$$1 \mapsto 3 \mapsto 5 \mapsto 1, \quad 2 \mapsto 4 \mapsto 2, \quad 6 \mapsto 6.$$

We can write these orbits in cyclic notation as $(1\ 3\ 5)$, $(2\ 4)$ and (6). Putting them all together allows us to write

$$\sigma = (1\ 3\ 5) \circ (2\ 4) \circ (6), \tag{2.3}$$

where we sometimes drop the 'follows' symbols, on the understanding that we read the cycles from right to left, giving $\sigma = (1\ 3\ 5)(2\ 4)(6)$.

We now generalise the description of σ from (2.3). For $\sigma \in S_n$ and $r \in \mathbb{N}$, write

$$\sigma^r = \underbrace{\sigma \circ \cdots \circ \sigma}_{r}$$

for the composition of σ with itself taken r times. Define $\sigma^0 = \mathrm{id}_{\{1,\ldots,n\}}$, and for negative integers, write $\sigma^{-r} = (\sigma^{-1})^r$.

Lemma 2.68. *Let $n \in \mathbb{N}$ and consider $\sigma \in S_n$. The relation $\sim$ on $\{1, \ldots, n\}$ defined by*

$$a \sim b \iff \sigma^r(a) = b \ \text{ for some } \ r \in \mathbb{Z}$$

is an equivalence relation. Note that $\sim$ depends on σ. For $a \in \{1, \ldots, n\}$, the equivalence class containing a, called the orbit of a, *is of the form $\{a, \sigma(a), \ldots, \sigma^{m-1}(a)\}$ for some $m \in \mathbb{N}$.*

Proof. First, note that each $a \in \{1, \ldots, n\}$ satisfies $a = \mathrm{id}_{\{1,\ldots,n\}}(a) = \sigma^0(a)$. Thus $a \sim a$, so $\sim$ is reflexive. Next, suppose that $a \sim b$. Then $b = \sigma^r(a)$ for some $r \in \mathbb{Z}$ and hence $a = \sigma^{-r}(b)$, so $b \sim a$. Finally, suppose that $a \sim b$ and $b \sim c$. Then there exist $r, s \in \mathbb{Z}$ such that $b = \sigma^r(a)$ and $c = \sigma^s(b)$. Then $c = \sigma^r(\sigma^s(a)) = \sigma^{r+s}(a)$ and hence $a \sim c$.

For the second statement, the orbit of a comprises all elements $\sigma^r(a)$ for $r \in \mathbb{Z}$. Since $\{1, \ldots, n\}$ is a finite set, there are only finitely many distinct elements of the form $\sigma^r(a)$, and hence there exists $r, s \in \mathbb{N}$ with $r < s$ such that $\sigma^r(a) = \sigma^s(a)$. It follows that $\sigma^{s-r}(a) = a$. Let m be the smallest positive integer such that $\sigma^m(a) = a$, so the orbit containing a is

$$\{a, \sigma(a), \ldots, \sigma^{m-1}(a)\}$$

as required. Note in passing that these elements are distinct: if $\sigma^r(a) = \sigma^s(a)$ and $0 \le r < s < m$, then $\sigma^{s-r}(a) = a$ and $0 < s - r < m$, contradicting minimality of m. $\square$

It now follows from Theorem 2.63 that $\{1, \ldots, n\}$ is partioned into the (necessarily non-empty, disjoint) orbits, where σ cyclically permutes the elements in each orbit, i.e.

$$a \mapsto \sigma(a) \mapsto \ldots \mapsto \sigma^{m-1}(a) \mapsto a. \tag{2.4}$$

for some $m \leq n$ that depends on the choice of a.

Definition 2.69 (Cyclic permutations and transpositions). For $m < n$, an *m-cycle* in S_n is a permutation $\sigma \in S_n$ that cyclically permutes m elements of $\{1, \ldots, n\}$ as in (2.4) and fixes the remaining elements. We write $\sigma = \big(a\ \sigma(a)\ \ldots\ \sigma^{m-1}(a)\big)$. A 2-cycle is called a *transposition*.

Suppose that some $\sigma \in S_n$ has k orbits, say $O_1, \ldots, O_k$ of sizes $m_1, \ldots, m_k$. For each orbit O_i, let σ_i be the m_i-cycle that permutes the elements in O_i cyclically in the same way as σ but fixes the remaining elements. Then σ decomposes as the composition

$$\sigma = \sigma_1 \sigma_2 \cdots \sigma_k,$$

just as in (2.3). In fact, we can go further.

Lemma 2.70. *Every permutation in S_n can be written as a composition of transpositions.*

Proof. We have seen above that every permutation can be written as a composition of cycles, so we need only consider these. Notice that

$$(1\ 2\ \ldots\ r) = (1\ r)(1\ r-1) \cdots (1\ 3)(1\ 2),$$

and more generally, notice that

$$(i_1\ i_2\ \ldots\ i_r) \ = \ (i_1\ i_r)(i_1\ i_{r-1}) \cdots (i_1\ i_2)$$

as required. □

Example 2.71. For the permutation σ from Example 2.67, we have already written

$$\alpha = (1\ 3\ 5)(2\ 4)(6) = (1\ 3\ 5)(2\ 4) = (1\ 5)(1\ 3)(2\ 4).$$

We might equally well have written $\alpha = (1\,5)(1\,3)(2\,4)(1\,2)(1\,2) = (1\,3)(1\,5)(1\,3)(1\,5)(2\,4)$.

In this example, notice that in all three decompositions of σ as a product of transpositions, the number of transpositions is odd. We saw in Lemma 2.70 that an r-cycle can be written as the product of composition of $r - 1$ transpositions. It turns out that the parity of the number of transpositions in any permutation σ is intrinsic to σ.

Definition 2.72 (Even and odd permutations). A permutation is *odd* (resp. *even*) if it can be written as the composition of an odd (resp. *even*) number of transpositions.

It is not clear at this stage that this is well-defined: on the face of it, one decomposition might comprise an odd number of transpositions, while another one might comprise an even number. In order to help us justify the definition, consider the following function.

Definition 2.73 (**Sign of a permutation**)**.** Let sign: $S_n \to \{1, -1\}$ be the function defined by sending a permutation σ whose k orbits are cycles of lengths $m_1, \ldots, m_k$ to

$$\mathrm{sign}(\sigma) = (-1)^{(m_1-1)+(m_2-1)+\cdots+(m_k-1)}.$$

Example 2.74. The permutation $\sigma = (1\ 3\ 5) \circ (2\ 4) \circ (6)$ from Example 2.67 has three orbits of sizes 3, 2 and 1, so

$$\mathrm{sign}(\sigma) = (-1)^{(3-1)+(2-1)+(1-1)} = -1.$$

Notice that the orbit of size $m = 1$ does not contribute to to the value of $\mathrm{sign}(\sigma)$ as $m - 1 = 0$.

Lemma 2.75. *Let $\sigma \in S_n$ and $(i\ j)$ be a transposition in S_n. Then*

$$\mathrm{sign}\big((i\ j) \circ \sigma\big) = (-1) \cdot \mathrm{sign}(\sigma).$$

Proof. (Non-examinable) Suppose σ has k orbits $O_1, \ldots, O_k$ of lengths $m_1, \ldots, m_k$. Then:

Case 1. Suppose first that i and j belong to the same orbit which, after reordering if necessary, we may write as O_1. Write this orbit starting at i, and let r denote the number of places separating j from i in the cyclic order, so the orbit is $(i\ i_1\ \ldots\ i_r\ j\ j_1\ \ldots\ j_s)$ for s satisfying $m_1 = r + s + 2$. One can calculate (check it!) that

$$(i\ j) \circ (i\ i_1\ \ldots\ i_r\ j\ j_1\ \ldots\ j_s) = (i\ i_1\ \ldots\ i_r) \circ (j\ j_1 \ldots\ j_s). \tag{2.5}$$

Therefore, the product $(i\ j) \circ O_1$ splits into two cycles of lengths $r + 1$ and $s + 1$, so the orbits of $(i\ j) \circ \sigma$ are these two as well as the (unaffected) orbits $O_2, \ldots, O_k$. Hence

$$\begin{aligned} \mathrm{sign}\,\big((i\ j) \circ \sigma\big) &= (-1)^{r+s+(m_2-1)+\cdots+(m_k-1)} \\ &= (-1) \cdot (-1)^{r+s+1+(m_2-1)+\cdots+(m_k-1)} \\ &= (-1) \cdot (-1)^{(m_1-1)+(m_2-1)+\cdots+(m_k-1)} \\ &= (-1) \cdot \mathrm{sign}(\sigma). \end{aligned}$$

Case 2. Otherwise, i and j belong to different orbits, say O_1 and O_2 respectively, so we may write the corresponding cycles for these orbits as $(i\ i_1\ \ldots\ i_r)$ and $(j\ j_1\ \ldots\ j_s)$. By taking the composition with $(i\ j)$ on both sides of (2.5) from the left, we have

$$(i\ j) \circ (i\ i_1\ \ldots\ i_r) \circ (j\ j_1\ \ldots\ j_s) = (i\ i_1\ \ldots\ i_r\ j\ j_1\ \ldots\ j_s).$$

Thus, in this case, the product $(i\ j) \circ O_1 O_2$ merge into a single orbit $r + s + 2$. Just as in Case 1, we obtain $\mathrm{sign}\big((i\ j) \circ \sigma\big) = (-1) \cdot \mathrm{sign}(\sigma)$.

Thus, in either case, we conclude that $\mathrm{sign}\big((i\ j) \circ \sigma\big) = (-1) \cdot \mathrm{sign}(\sigma)$ as required. □

Corollary 2.76. *A permutation σ is even or odd if and only if* $\mathrm{sign}(\sigma)$ *equals* 1 *or* -1 *respectively.*

Proof. (Non-examinable) Using $\mathrm{sign}\big((i\ j)\big) = (-1)^{2-1} = (-1)$, we see by repeated use of Lemma 2.75 that

$$\mathrm{sign}\big((i_1\ j_1) \cdots (i_r\ j_r)\big) = (-1)^r,$$

that is, -1 if r is odd and 1 if r is even. □

3 Polynomials and Unique Factorisation

We saw in Section 1.3 that the study of $\mathbb{Z}$, equipped with the usual operations of addition and multiplication, leads to some powerful results, including the Fundamental Theorem of Arithmetic (see Theorem 1.33). It turns out that there is a strong parallel between this study of $\mathbb{Z}$ on one hand, and on the other, the study of the set of polynomials with coefficients in a field.

3.1 On fields and rings

We want to choose from a specified 'field of scalars' such as $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{Z}_p$ for p prime. Having coefficients in a field will make the theory simpler at certain crucial points. The definition is rather long...

Definition 3.1 (**Fields and rings**). A *field* is a set $\mathbb{F}$ that comes equipped with two binary operations that we call addition (or sum), namely

$$\text{for every } x, y \in \mathbb{F}, \text{ we have } x + y \in \mathbb{F},$$

and multiplication (or product), namely

$$\text{for every } x, y \in \mathbb{F}, \text{ we have } xy \in \mathbb{F},$$

such that the following (extended) field axioms hold:

F1 $+$ is associative: $\forall x, y, z \in \mathbb{F}$, $(x + y) + z = x + (y + z)$; so we can omit brackets.

F2 There is an additive identity $0 \in \mathbb{F}$ s.t. $0 + x = x = x + 0$.

F3 Every $x \in \mathbb{F}$ has an additive inverse $-x \in \mathbb{F}$ such that $x + (-x) = 0 = -x + x$; so we also have subtraction $x - y := x + (-y)$.

F3x $\forall x \in \mathbb{F}$, we have $-(-x) = x$

F4 $+$ is commutative: $\forall x, y \in \mathbb{F}$ $x + y = y + x$.

F5 $\cdot$ is distributive over $+$: $\forall a, x, y \in \mathbb{F}$

$$a \cdot (x + y) = a \cdot x + a \cdot y$$
$$(x + y) \cdot a = x \cdot a + y \cdot a$$

F5x $\forall a, x \in \mathbb{F}$

$$a \cdot 0 = 0 = 0 \cdot a$$
$$a \cdot (-x) = -(a \cdot x)$$
$$(-x) \cdot a = -(x \cdot a)$$

F6 $\cdot$ is associative $\forall x, y, z \in \mathbb{F}$ $(x \cdot y) \cdot z = x \cdot (y \cdot z)$

F7 There is a multiplicative identity $1 \in \mathbb{F}$, $\forall x \in \mathbb{F}$ $1 \cdot x = x = x \cdot 1$.

F8 $\cdot$ is commutative $\forall x, y \in \mathbb{F}$ $x \cdot y = y \cdot x$

F9 $1 \neq 0$ and $\forall x \in \mathbb{F} \setminus \{0\}$ there is a multiplicative inverse x^{-1}, s.t. $x^{-1} \cdot x = 1 = x \cdot x^{-1}$.

F9x $\forall x \in \mathbb{F} \setminus \{0\}$, $(x^{-1})^{-1} = x$.

Remark 3.2. It's worth also introducing a little more terminology.

1. With just (F1)-(F4), we say $\mathbb{F}$ (with $+, 0$) is an *additive group*; in particular we can unambiguously calculate sums like $\sum_{j=1}^{n} x_j$ for $x_j \in \mathbb{F}$.
2. If (F1)-(F7) hold, we say $\mathbb{F}$ is a *ring* (some add 'with 1' for emphasis).
3. If (F1)-(F8) hold, we say $\mathbb{F}$ is a *commutative ring*.

Example 3.3. The sets $\mathbb{Z}$ and $\mathbb{Z}_n$, for n not prime, with the usual addition and multiplication are commutative rings. However, they are not fields as (F9) does not hold.

Definition 3.4 (Integral domains). A commutative ring $\mathbb{F}$ with $0 \neq 1$ is an *integral domain* if, $\forall\, x, y \in \mathbb{F}$, we have that $xy = 0 \implies x = 0$ or $y = 0$.

Lemma 3.5 (The cancellation law). *A commutative ring $\mathbb{F}$ is an integral domain if and only if $\forall x, y, z \in \mathbb{F}$, we have that*

$$(xy = xz \text{ and } x \neq 0) \implies y = z. \tag{3.1}$$

Proof. The characterising property of an integral domain is logically equivalent to the condition that

$$\forall x, y \in \mathbb{F}, \text{if } xy = 0 \text{ and } x \neq 0, \text{ then } y = 0. \tag{3.2}$$

To prove the lemma, we need only show that the cancellation law (3.1) holds if and only if (3.2) holds.

First, suppose (3.1) holds. The special case $z = 0$ is precisely the statement of (3.2). For the converse, suppose that (3.2) holds and that $xy = xz$ with $x \neq 0$. Axiom (F3) gives that $xy + (-xz) = 0$ which, by (F5x), is $xy - xz = 0$. Now (F5) gives $x(y - z) = 0$, and since $x \neq 0$, assumption (3.2) implies that $y - z = 0$. Apply (F5x) and then (F1) and (F2) to obtain $y = z$ as required. □

Remark 3.6. The cancellation law holds in any field: multiply $xy = xz$ by x^{-1} and use (F6, 7, 9). However, the cancellation law also holds in $\mathbb{Z}$, so it is weaker than (F9). It does not hold in $\mathbb{Z}_n$ for n non-prime. For example, in $\mathbb{Z}_6$, we have $[2] \cdot [3] = [0]$ and of course $[2] \neq [0]$, but we also have $[3] \neq [0]$. This shows in particular that the commutative ring $\mathbb{Z}_6$ is not an integral domain.

3.2 The polynomial ring

Let $\mathbb{F}$ be any field.

Definition 3.7 (Polynomials). A *polynomial* in X with coefficients in $\mathbb{F}$ is an expression

$$f = f(X) = \sum_{n=0}^{\infty} a_n X^n,$$

where the coefficients $a_n \in \mathbb{F}$, and where all but finitely many of the a_n are zero. Two polynomials are equal if and only if all of their coefficients are equal. Define addition and multiplication of polynomials as follows: if also $g = \sum_{n=0}^{\infty} b_n X^n$, then

$$f + g = \sum_{n=0}^{\infty}(a_n + b_n)X^n, \qquad f \cdot g = \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} a_k b_{n-k}\right) X^n.$$

We write $\mathbb{F}[X]$ for the resulting *polynomial ring with coefficients in* $\mathbb{F}$.

Remark 3.8.

1. The constant polynomials 0 and 1 are the additive and multiplicative identities in $\mathbb{F}[X]$. Also $-f$ is the polynomial with coefficients $-a_n$ for all $n \in \mathbb{N}_0$. In fact, $\mathbb{F}[X]$ is a commutative ring, but it is not a field. The only invertible polynomials are the non-zero constants, denoted $\mathbb{F}^* = \mathbb{F} \setminus \{0\}$.

2. In practice, we simplify this sum using standard rules

$$X^0 = 1, \quad X^1 = X, \quad 1 \cdot X^n = X^n, \quad 0 \cdot X^n = 0$$

and we omit all large powers with coefficients $a_n = 0$, so that, for example,

$$f = 2 + X + 5X^3 \quad \text{in place of} \quad 2X^0 + 1 \cdot X^1 + 0 \cdot X^2 + 5X^3 + 0 \cdot X^4 + \cdots,$$

with $a_n = 0$, for $n \geq 4$.

3. Every $a \in \mathbb{F}$ determines a constant polynomial with $a_0 = a$ and $a_n = 0$ for $n > 0$. These add and multiply as in $\mathbb{F}$.

Example 3.9. The operations of addition and multiplication in $\mathbb{F}[X]$ are both very familiar: $(X + 1) + (X + 2) = 2X + 3$, while $(X + 1)(X + 2) = X^2 + 3X + 2$.

Remark 3.10. A polynomial $f \in \mathbb{F}[X]$ determines an 'evaluation' function

$$\mathrm{ev}_f \colon \mathbb{F} \to \mathbb{F} \colon \alpha \mapsto f(\alpha) = \sum_{n=0}^{\infty} a_n \alpha^n$$

that is, setting $X = \alpha$. However, it turns out that one shouldn't always consider f to be equal to this function. This is a very subtle point and it depends very much on the field $\mathbb{F}$, and in fact, to make the point we have to use the finite field $\mathbb{F} = \mathbb{Z}_p$. In this case, the set of functions $\mathbb{F}^{\mathbb{F}}$ is finite of size p^p, but $\mathbb{F}[X]$ is always infinite. For example, $1, X, X^2, X^3, \ldots$ are infinitely many different polynomials, but, by Fermat's Little Theorem, we know that $\alpha^p = \alpha$ for all $\alpha \in \mathbb{Z}_p$, meaning that, as functions, $\mathrm{ev}_{X^p} = \mathrm{ev}_X$.

Definition 3.11 (Leading coefficient). For any non-zero $f \in \mathbb{F}[X]$, the *leading term* is $a_k X^k$ if $a_k \neq 0$ and $a_n = 0$ for $n > k$, in which case the *leading coefficient* is a_k and k is the *degree*[11] of f, written $\deg f = k$. We say f is *monic* if the leading coefficient $a_k = 1$.

Remarks 3.12.
1. Notice that the constant polynomials are those of degree equal to $-\infty$ or 0; equivalently, the non-constant polynomials are those of degree ≥ 1.
2. For any non-zero f, with leading coefficient a, the polynomial $p = a^{-1}f$ is monic and $f = ap$. For example, $f = 2X^2 - 3X + 1 = 2(X^2 - \frac{3}{2}X + \frac{1}{2})$. Thus every non-zero polynomial can be written (uniquely) as a non-zero constant times a monic polynomial. We can view this as parallel to the fact that every non-zero integer can be written uniquely as ± 1 times a positive integer.

Lemma 3.13. *Let $f, g \in \mathbb{F}[X]$. The degree satisfies*

$$\deg(fg) = \deg(f) + \deg(g). \tag{3.3}$$

Moreover, if $fg = 0$, then $f = 0$ or $g = 0$, i.e. Lemma 3.5 holds in $\mathbb{F}[X]$.

Proof. The footnote in our definition of degree deals with the case where either f or g is zero, so we may assume that neither is zero. Then we may write

$$\begin{aligned} f &= a_0 + a_1 X + \ldots + a_m X^m, && \text{with } a_m \neq 0 \\ g &= b_0 + b_1 X + \ldots + b_n X^n, && \text{with } b_n \neq 0 \end{aligned}$$

Thus $\deg f = m$ and $\deg g = n$ and the leading term of fg is $a_m b_n X^{n+m}$. Now $a_m b_n \neq 0$, because the cancellation law holds in $\mathbb{F}$, so $fg \neq 0$, because it has a non-zero coefficient, and hence $\deg(fg) = m + n$ as claimed. □

3.3 Unique factorisation for polynomials

You may find it very helpful to compare the next definition to Definition 1.19.

Definition 3.14 (Irreducible polynomials). Let $f, g \in \mathbb{F}[X]$. We say that f *divides* g, written $f \mid g$, if there exists $h \in \mathbb{F}[X]$ such that $fh = g$. Given a nonzero polynomial $p \in \mathbb{F}[X]$, we say that p is *irreducible* if p is not constant and if any factorisation $p = fg$ forces either f or g to be constant.

Remarks 3.15.
1. Equivalently, $p \in \mathbb{F}[X]$ is irreducible iff $\deg p > 0$ and we cannot write $p = fg$ with $\deg f > 0$ and $\deg g > 0$. Compare Definition 1.19, which says $p \in \mathbb{Z}$ is prime iff $p \geq 2$ and we cannot write $p = ab$ with $a \geq 2$ and $b \geq 2$.
2. If $f, g \in \mathbb{F}[X]$ with $g \neq 0$, then $f \mid g$ implies that $\deg(f) \leq \deg(g)$. This follows from (3.3) because if $g = fh$, the $\deg(g) = \deg(f) + \deg(h) \geq \deg(f)$ as $h \neq 0$.

[11] By convention, the degree of the zero polynomial is defined to be $\deg(0) = -\infty < n$ for any $n \in \mathbb{N}_0$, and we define this to satisfy $-\infty + n = -\infty$ for all $n \in \mathbb{N}$, and $-\infty - \infty = -\infty$.

3. For all $f \in \mathbb{F}[X]$, $1 \mid f$ and $f \mid 0$. However, $0 \mid f \implies f = 0$, and $f \mid 1 \implies f$ is a nonzero constant.

Let's now formulate and prove some statements that need a little more justification.

Lemma 3.16. *Let $f, g, h \in \mathbb{F}[X]$.*

(1) If $f \mid g$ and $g \mid f$, then $g = cf$ for some nonzero constant $c \in \mathbb{F}$.

(2) If $h \mid f$ and $h \mid g$, then $h \mid \lambda f + \mu g$ for all $\lambda, \mu \in \mathbb{F}[X]$.

Proof. For (1), there exists $c, d \in \mathbb{F}[X]$ such that $cf = g$ and $dg = f$. Substituting the first equation into the second gives $cdf = f$. If $f = 0$, then $g = 0$, and we may choose any constant c. Otherwise $f \neq 0$, in which case the cancellation law gives $cd = 1$. Therefore $\deg(c) + \deg(d) = \deg(1) = 0$, so $\deg(c) = \deg(d) = 0$ and hence c, d are non-zero constant polynomials. For (2), there exists $x, y \in \mathbb{F}[X]$ such that $hx = f$ and $hy = g$. Then for any $\lambda, \mu \in \mathbb{F}[X]$, we have $\lambda f + \mu g = h(\lambda x + \mu y)$, so $h \mid \lambda f + \mu g$. $\square$

We could now state and prove a factorisation result for $\mathbb{F}[X]$ just like the statement of Proposition 1.22 for $\mathbb{Z}$. However, just as we strengthened that result to the Fundamental Theorem of Arithmetic (which is existence *and* uniqueness of a decomposition), we now study $\mathbb{F}[X]$ further in order to state the analogue of Theorem 1.33 for $\mathbb{F}[X]$.

Definition 3.17 (lcm and gcd). Let $f, g \in \mathbb{F}[X]$, we say that $h \in \mathbb{F}[X]$ is:

- a *common multiple* of f and g if $f \mid h$ and $g \mid h$. It is a *least common multiple (lcm) of f and g* if

$$f \mid h \text{ and } g \mid h \text{ and } \forall k \in \mathbb{F}[X], \text{ if } f \mid k \text{ and } g \mid k, \text{ then } h \mid k.$$

- a *common divisor* of f and g if $h|f$ and $h|g$. It is a *greatest common divisor (gcd) of f and g* if

$$h \mid f \text{ and } h \mid g \text{ and } \forall k \in \mathbb{F}[X], \text{ if } k \mid f \text{ and } k \mid g, \text{ then } k \mid h.$$

We say that f, g are *coprime* if 1 is a gcd of f, g, that is, $x \mid f$ and $x \mid g \iff x \mid 1$.

Remark 3.18. We have not yet shown that gcd's and lcm's exist (see Proposition 3.21 and Theorem 3.25).

Proposition 3.19 (Division for Polynomials). *Let $f, g \in \mathbb{F}[X]$ with $g \neq 0$. There are unique $q, r \in \mathbb{F}[X]$ such that $f = qg + r$ and $\deg r < \deg g$ (including $r = 0$).*

Proof. If $f = 0$, then $q = r = 0$ is the unique solution. Otherwise, $\deg(f) \geq 0$. If $\deg f < \deg g$, then $q = 0$, $r = f$ is the unique solution (otherwise $q \neq 0$, in which case $\deg(f) = \deg(qg + r) = \deg(qg) \geq \deg(g) > \deg(f)$ which is absurd). For $\deg f \geq \deg g$ proceed by induction on $\deg f$. Suppose f and g have leading terms aX^m and bX^n

respectively, so a, b are non-zero and $m \geq n$. As the coefficients are in a field $\mathbb{F}$, b is invertible, so we can consider

$$f_1 = f - \frac{a}{b} X^{m-n} \cdot g,$$

so the leading terms on the RHS cancel and $\deg f_1 < \deg f$. Hence, by the inductive hypothesis, $f_1 = q_1 g + r$ uniquely with $\deg r < \deg g$, and so

$$f = f_1 + \frac{a}{b} X^{m-n} g = \left(\frac{a}{b} X^{m-n} + q_1 \right) g + r,$$

giving $q = \frac{a}{b} X^{m-n} + q_1$, as required. $\square$

Note that the proof gives a recursive method for calculation, and this can be done in one line, matching each degree in turn. For example, if $f = X^4 + 3$ and $g = 2X^2 + X + 2$, then

$$X^4 + 3 = (2X^2 + X + 2) \left(\frac{1}{2} X^2 - \frac{1}{4} X - \frac{3}{8} \right) + \frac{7}{8} X + 3\frac{3}{4}$$

Notice that we do need to allow coefficients to be in a field ($\mathbb{Q}$ in this case) for the computation to work.

Corollary 3.20 (**Characterising roots of polynomials**). *If $f \in \mathbb{F}[X]$ and $\alpha \in \mathbb{F}$, then*

$$f = (X - \alpha) q + f(\alpha) \quad \text{for some } q \in \mathbb{F}[X]$$

Hence, α is a root of f, that is $f(\alpha) = 0$, if and only if $(X - \alpha) \mid f$.

Proof. By the Division Proposition 3.19 with $g = X - \alpha$, we obtain $f = (X - \alpha) q + r$ for some $q, r \in \mathbb{F}[X]$ with $\deg r < \deg(X - \alpha) = 1$. Hence $r \in \mathbb{F}$ is a constant polynomial and we see that $r = f(\alpha)$ by evaluating at α. The last part then follows immediately. $\square$

Proposition 3.21 (**Bézout's identity for polynomials**). *For non-zero $f, g \in \mathbb{F}[X]$, let h be a non-zero polynomial of lowest degree in the set*

$$S = \{ \lambda f + \mu g \mid \lambda, \mu \in \mathbb{F}[X] \}.$$

Then h is a gcd *of f, g and every such gcd arises in this way.*

Proof. First, if $k \in \mathbb{F}[X]$ satisfies $k \mid f$ and $k \mid g$, then $k \mid h$ by Lemma 3.16(2) because $h \in S$. It therefore remains to prove that $h \mid f$ and $h \mid g$. Since $h \in S$, we may write $h = \lambda f + \mu g$ for some $\lambda, \mu \in \mathbb{F}[X]$. Division for polynomials gives $q, r \in \mathbb{F}[X]$ such that $f = hq + r$ with $\deg r < \deg h$. Then

$$r = f - hq = (1 - \lambda q) f - (\mu q) g \in S.$$

But, since h has lowest degree in S, we can only have $\deg r < \deg h$ if $r = 0$, i.e. $h \mid f$. Similarly, $h \mid g$ and so h is a gcd of f, g. To see that every gcd has this form, note that any other gcd is of the form ah for some $a \in \mathbb{F}^*$ by Lemma 3.16(1). Now, $ah = a\lambda f + a\mu g$, so $ah \in S$, while $\deg(ah) = \deg(h)$, so ah also has lowest degree (and is non-zero). $\square$

Corollary 3.22. *In* $\mathbb{F}[X]$*,* gcd*s exist and we can find them by Euclid's Algorithm.*

Example 3.23. To find the gcd of $f = X^9 - 1$ and $g = X^6 - 1$

$$\begin{aligned}(X^9 - 1) &= (X^6 - 1)X^3 + (X^3 - 1)\\ (X^6 - 1) &= (X^3 - 1)(X^3 + 1) + 0 \quad \text{STOP}\end{aligned}$$

Hence the (monic) gcd of $X^9 - 1$ and $X^6 - 1$ is $X^3 - 1$. In Euclid's Algorithm, it can be helpful to rescale the polynomials to be monic, since this does not change the gcd.

Proposition 3.24 (Euclid's Lemma for polynomials). *If* $f \in \mathbb{F}[X]$ *is irreducible, then*

$$f \mid gh \implies f \mid g \text{ or } f \mid h.$$

Proof. Suppose f divides gh and f does not divide g. Since f is irreducible, the common divisors of f, g are just $\mathbb{F}^*$, so 1 is a gcd. Hence we can write $\lambda f + \mu g = 1$, so $\lambda fh + \mu gh = h$ and so, as $f \mid gh$, we get $f \mid h$, as required. □

Theorem 3.25 (Unique Factorisation for polynomials). *Every non-constant polynomial can be written uniquely (upto the choice of order) as a product of irreducibles. In fact, each* $f \in \mathbb{F}[X]$ *of degree* $\deg(f) \geq 1$ *can be written as*

$$f = c\prod_{i=1}^{k} p_i = cp_1p_2\cdots p_k$$

for $k \geq 1$*, where* $c \in \mathbb{F}^*$ *is the leading coefficient of* f *and each* p_i *is monic and irreducible.*

Proof. (Sketch). We use the same argument as for $\mathbb{Z}$. For existence of the decomposition (compare the proof of Proposition 1.22), if f is irreducible, then $k = 1$ and we set $p_1 = f/c$ where c is the leading coefficient of f. Otherwise, $f = gh$ where $0 < \deg g, \deg h < \deg f$, so we obtain a factorisation of both g and h into irreducibles by induction on $\deg f$. This gives our desired factorisation. For uniqueness (compare the proof of Theorem 1.33), use Euclid's Lemma just as we did for $\mathbb{Z}$. □

Remark 3.26. This result allows us to define $\mathrm{lcm}(f, g)$ just as in Remark 1.34.

Remark 3.27. If our field $\mathbb{F}$ equals $\mathbb{C}$, then the Fundamental Theorem of Algebra (see Theorem 1.14) tells us that the unique factorisation of any $f \in \mathbb{C}[X]$ is

$$f = c\prod_{i=1}^{k}(X - \alpha_i)$$

where c is the leading coefficient of f and $\alpha_1, \ldots, \alpha_k$ are the roots of f. However, over $\mathbb{R}$ or $\mathbb{Q}$ or $\mathbb{Z}_p$ (or other fields), irreducible polynomials may not be linear (deg = 1). For example, in $\mathbb{R}[X]$, we have $X^3 - 1 = (X - 1)(X^2 + X + 1)$. Here both factors are irreducible, because $X^2 + X + 1$ already has two roots $(-1 \pm i\sqrt{3})/2$ in $\mathbb{C} \setminus \mathbb{R}$, so can have no roots in $\mathbb{R}$, as they would be extra roots in $\mathbb{C}$. Thus $X^2 + X + 1$ has no linear factors in $\mathbb{R}[X]$ (and it cannot have nontrivial factors of greater degree as it is only of degree two).

4 Matrices and Linear Systems

As in the previous section, let $\mathbb{F}$ denote a field.

4.1 The ring of matrices

Definition 4.1 (Matrix). An $m \times n$ *matrix with coefficients in a field* $\mathbb{F}$ is an array

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix}$$

with m rows and n columns and coefficients $a_{ij} \in \mathbb{F}$. We also write just $A = (a_{ij})$, when n, m are understood. Two such matrices are equal when all coefficients are equal, that is, if $B = (b_{ij})$, then $A = B$ when $a_{ij} = b_{ij}$ for all i, j. We write $\mathrm{M}_{m\times n}(\mathbb{F})$ for the set of all $m \times n$ matrices with coefficients in a field $\mathbb{F}$.

Matrices of the same size can be added and multiplied by a scalar $\lambda \in \mathbb{F}$ coefficient-wise:

$$A + B = (a_{ij} + b_{ij}), \qquad \lambda A = (\lambda a_{ij}).$$

For example,

$$\begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} + 5 \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 5 \\ 5 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 6 & 7 \end{pmatrix}.$$

Thus $\mathrm{M}_{m\times n}(\mathbb{F})$ is an additive group, as axioms (F1)-(F4) follow coefficient-wise. In particular the zero matrix $\mathbf{0}$ has all coefficients equal to 0. In fact, with scalar multiplication $\mathrm{M}_{m\times n}(\mathbb{F})$ is a 'vector space over $\mathbb{F}$' (see Algebra 1B for more on this).

The key operation is matrix multiplication: if we now have $B = (b_{jk}) \in \mathrm{M}_{n\times p}(\mathbb{F})$, then we can form $AB = (u_{ik}) \in \mathrm{M}_{m\times p}(\mathbb{F})$, where

$$u_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}.$$

For example,

$$\begin{pmatrix} r & s & t \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = rx + sy + tz \quad \text{or} \quad \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 3 \\ 4 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 8 \\ 16 & 9 \end{pmatrix}$$

Lemma 4.2. *Matrix multiplication is*

(1) associative: for $A \in \mathrm{M}_{k\times m}(\mathbb{F})$, $B \in \mathrm{M}_{m\times n}(\mathbb{F})$ and $C \in \mathrm{M}_{n\times p}(\mathbb{F})$,

$$(AB)C = A(BC);$$

(2) bilinear: for $A \in \mathrm{M}_{k\times m}(\mathbb{F})$, $B, C \in \mathrm{M}_{m\times n}(\mathbb{F})$, $D \in \mathrm{M}_{n\times p}(\mathbb{F})$ and $\lambda, \mu \in \mathbb{F}$,

$$\begin{aligned} A(\lambda B + \mu C) &= \lambda AB + \mu AC \\ (\lambda B + \mu C)D &= \lambda BD + \mu CD. \end{aligned}$$

Notice that setting $\lambda = \mu = 1$ recovers the distributive laws.

Proof. In each case, we need to show that the (i,j)-entries on both sides are equal.

$$\begin{aligned} (1) \qquad ((AB)C)_{ij} = \sum_{t=1}^{n} (AB)_{it} c_{tj} &= \sum_{t=1}^{n} \Big(\sum_{s=1}^{m} a_{is} b_{st} \Big) c_{tj} \\ &= \sum_{t=1}^{n} \sum_{s=1}^{m} a_{is} b_{st} c_{tj} \\ &= \sum_{s=1}^{m} a_{is} \sum_{t=1}^{n} b_{st} c_{tj} \\ &= (A(BC))_{ij}. \end{aligned}$$

$$\begin{aligned} (2) \qquad (A(\lambda B + \mu C))_{ij} = \sum_{s=1}^{m} a_{is}(\lambda b_{sj} + \mu c_{sj}) &= \lambda \sum_{s=1}^{m} a_{is} b_{sj} + \mu \sum_{s=1}^{m} a_{is} c_{sj} \\ &= (\lambda AB)_{ij} + (\mu AC)_{ij}. \end{aligned}$$

The proof of the other identity is similar. We have used the fact that multiplication in $\mathbb{F}$ is associative, distributive and, for (2), commutative. □

Remark 4.3. The set $\mathrm{M}_n(\mathbb{F}) := \mathrm{M}_{n\times n}(\mathbb{F})$ of $n \times n$ (square) matrices over $\mathbb{F}$ forms a ring under matrix addition and multiplication. We have already noted (F1)-(F4), while (F5) (distributive) and (F6) (associative) follow from Lemma 4.2. The multiplicative identity $\mathbb{I}_n \in \mathrm{M}_n(\mathbb{F})$ is the matrix

$$\mathbb{I} = (\delta_{ij}), \text{ where } \delta_{ij} = \begin{cases} 1 & i = j \text{ (on diagonal)} \\ 0 & i \neq j \text{ (off diagonal)} \end{cases}$$

We write $\mathbb{I}_n$ when we wish to specify the size, e.g.

$$\mathbb{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Check: $\mathbb{I}_n A = A = A\,\mathbb{I}_n$, for all $A \in \mathrm{M}_n(\mathbb{F})$. More generally, if $B \in \mathrm{M}_{n\times p}(\mathbb{F})$, then $\mathbb{I}_n B = B$ and, if $C \in \mathrm{M}_{k\times n}(\mathbb{F})$, then $C\,\mathbb{I}_n = C$.

Remark 4.4. The ring $\mathrm{M}_1(\mathbb{F}) = \mathbb{F}$ is in fact a field (because $\mathbb{F}$ is!), but for $n > 1$, the ring $\mathrm{M}_n(\mathbb{F})$ is not commutative and does not satisfy the cancellation law, e.g.

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

but

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Thus we have found $A, B \in \mathrm{Mat}_2(\mathbb{F})$, both non-zero, with $BA = \mathbf{0}$ and with $BA \neq AB$.

Definition 4.5 (Invertible matrix). A matrix $A \in \mathrm{M}_n(\mathbb{F})$ is *invertible* iff there exists $X \in \mathrm{M}_n(\mathbb{F})$ such that

$$XA = \mathbb{I}_n = AX \tag{4.1}$$

We will see that X is unique and so we can denote it by A^{-1} and call it the *inverse* of A.

Examples 4.6. 1. $\mathbb{I}_n$ is invertible and $\mathbb{I}_n^{-1} = \mathbb{I}_n$, because $\mathbb{I}_n\mathbb{I}_n = \mathbb{I}_n$.

2. A diagonal matrix A (i.e. $a_{ij} = 0$ if $i \neq j$) [(2)] with non-zero diagonal entries is simple to invert, e.g.

$$\begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}^{-1} = \begin{pmatrix} 3^{-1} & 0 \\ 0 & 4^{-1} \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}$$

In fact, the inverse of A is unique in the following stronger sense.

Lemma 4.7. *For $A \in \mathrm{M}_n(\mathbb{F})$, suppose that there are $X', X'' \in M_n(\mathbb{F})$ with $X'A = \mathbb{I}_n$ and $AX'' = \mathbb{I}_n$. Then $X' = X''$. In other words, if A has both a left inverse and a right inverse, then these must be equal and A is invertible.*

Proof. We repeat the argument from the proof of Theorem 2.37(3) to see that

$$X' = X'\mathbb{I}_n = X'(AX'') = (X'A)X'' = \mathbb{I}_nX'' = X'',$$

as required. □

Lemma 4.8. *(1) If $A \in \mathrm{M}_n(\mathbb{F})$ is invertible, then so is A^{-1}, and $(A^{-1})^{-1} = A$.*

(2) If $A, B \in \mathrm{M}_n(\mathbb{F})$ are invertible, then so is AB and $(AB)^{-1} = B^{-1}A^{-1}$.

Proof. See Exercise Sheet. □

Remark 4.9 (An aside). It follows that the set of invertible matrices in $\mathrm{M}_n(\mathbb{F})$ forms a group, known as the general linear group, denoted $\mathrm{GL}_n(\mathbb{F})$.

Example 4.10. [**Inverting a** 2×2 **matrix**] For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{M}_2(\mathbb{F}), \qquad \text{define} \qquad A' = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Then

$$AA' = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} = (\det A)\mathbb{I}_2$$

where we define the *determinant* $\det A = ad - bc \in \mathbb{F}$. Check that also $A'A = (\det A)\mathbb{I}_2$. Thus

$$A^{-1} = \frac{1}{\det A} A'$$

provided $\det A \neq 0$, so that $(\det A)^{-1}$ exists in $\mathbb{F}$ (as $\mathbb{F}$ is a field). You can also check, at least for determinants of 2×2 matrices, that $\det(AB) = (\det A)(\det B)$.

Remark 4.11 (**Another non-examinable aside**)**.** Determinants may be defined for all $n \times n$ matrices, and you'll see in Algebra 1B that A invertible $\iff$ $\det A \neq 0$, and moreover, that $\det(AB) = (\det A)(\det B)$ for any $A, B \in \mathrm{M}_n(\mathbb{F})$.

4.2 Matrices as linear maps

To any field $\mathbb{F}$, we associate the space of *coordinate vectors* $\mathbb{F}^n = \underbrace{\mathbb{F} \times \ldots \times \mathbb{F}}_{n \text{ times}}$.

This is an additive group with component wise operations

$$\begin{aligned} (x_1, \ldots, x_n) + (y_1, \ldots, y_n) &= (x_1 + y_1, \ldots, x_n + y_n) \\ -(x_1, \ldots, x_n) &= (-x_1, \ldots, -x_n) \\ \mathbf{0} &= (0, \ldots, 0) \end{aligned}$$

and the operation of scalar multiplication by $\lambda \in \mathbb{F}$

$$\lambda(x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n)$$

making $\mathbb{F}^n$ the prototype vector space (see Algebra 1B for an axiomatic treatment). For example, in $\mathbb{R}^2$

$$2(1,1) + 3(1,0) = (2,2) + (3,0) = (5,2).$$

and we can also take longer linear combinations like $\displaystyle\sum_{i=1}^{k} \lambda_i \mathbf{x}_i$ for $\mathbf{x}_i \in \mathbb{F}^n$ and $\lambda_i \in \mathbb{F}$.

We will identify $\mathbb{F}^n$ with $\mathrm{M}_{n\times 1}(\mathbb{F})$, the set of $n \times 1$ matrices or *column vectors*, in the natural way:

$$v = (v_1, \ldots, v_n) \in \mathbb{F}^n \text{ becomes } v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$$

and this respects the vector space structure on both sides.

Definition 4.12 (The linear map associated to a matrix). A matrix $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ determines a map

$$\phi_A \colon \mathbb{F}^n \to \mathbb{F}^m \colon \mathbf{x} \mapsto A\mathbf{x}$$

by matrix multiplication; specifically

$$A\mathbf{x} = \sum_{j=1}^{n} x_j \mathsf{col}_j(A),$$

is a linear combination of the columns of A, where $\mathsf{col}_j(A) \in \mathbb{F}^m$ is the jth column of A.

Example 4.13. The identity matrix satisfies $\phi_{\mathbb{I}_n} = \mathrm{id}_{\mathbb{F}^n} \colon \mathbb{F}^n \to \mathbb{F}^n$, because $\mathbb{I}_n\mathbf{x} = \mathbf{x}$.

Example 4.14. The matrix $A = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} \in \mathrm{M}_2(\mathbb{R})$ gives a map

$$\phi_A \colon \mathbb{R}^2 \to \mathbb{R}^2 \colon \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_1x + b_1y \\ a_2x + b_2y \end{pmatrix} = x \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + y \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}.$$

In other words,

$$\phi_A(x\mathbf{e}_1 + y\mathbf{e}_2) = x\phi_A(\mathbf{e}_1) + y\phi_A(\mathbf{e}_2),$$

where $\mathbf{e}_1 = (1,0)$ and $\mathbf{e}_2 = (0,1)$ are the standard unit vectors.

Note that ϕ_A determines A as its columns are $\phi_A(\mathbf{e}_1) = \mathbf{a}$ and $\phi_A(\mathbf{e}_2) = \mathbf{b}$.

Definition 4.15 (Linear map). A map $\psi \colon \mathbb{F}^n \to \mathbb{F}^m$ is *linear* if $\forall\, \mathbf{v}, \mathbf{w} \in \mathbb{F}^n$, $\lambda, \mu \in \mathbb{F}$,

$$\psi(\lambda\mathbf{v} + \mu\mathbf{w}) = \lambda\psi(\mathbf{v}) + \mu\psi(\mathbf{w}).$$

Remark 4.16. If ψ is linear, then

$$\psi\left(\sum_{i=1}^{k} \lambda_i \mathbf{v}_i\right) = \sum_{i=1}^{k} \lambda_i \psi(\mathbf{v}_i),$$

that is, ψ preserves arbitrary linear combinations.

Proposition 4.17. *Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $B \in \mathrm{M}_{n\times p}(\mathbb{F})$ and $C \in \mathrm{M}_n(\mathbb{F})$. Then:*

(1) the multiplication map $\phi_A \colon \mathbb{F}^n \to \mathbb{F}^m$ is linear;

(2) matrix multiplication is compatible with composition of linear maps, namely

$$\phi_{AB} = \phi_A \circ \phi_B \colon \mathbb{F}^p \to \mathbb{F}^m.$$

(3) if C is an invertible matrix with (two-sided) inverse X, then ϕ_C is a bijection with (two-sided) inverse ϕ_X.

Proof. For (1), we need to check that $A(\lambda\mathbf{x}+\mu\mathbf{y}) = \lambda A\mathbf{x} + \mu B\mathbf{y}$, which is a special case of the bilinearity of matrix multiplication from Lemma 4.2(2). For (2), we need to check that $A(B\mathbf{x}) = (AB)\mathbf{x}$, which is a special case of the associativity of matrix multiplication from Lemma 4.2(1). For (3), since C is invertible, we know C^{-1} exists. Applying part (2) to the matrix products CC^{-1} and $C^{-1}C$ gives that $\phi_{C^{-1}} \circ \phi_C = \phi_{C^{-1}C} = \phi_{\mathbb{I}_n} = \mathrm{id}_{\mathbb{F}^n}$ and $\phi_C \circ \phi_{C^{-1}} = \phi_{CC^{-1}} = \mathrm{id}_{\mathbb{F}^n}$, so ϕ_C is a bijection with ${\phi_C}^{-1} = \phi_{C^{-1}}$. $\square$

Since matrix multiplication corresponds to composition of maps, the associativity of composition means

$$\begin{aligned}\phi_{(AB)C} = \phi_{AB} \circ \phi_C &= (\phi_A \circ \phi_B) \circ \phi_C \\ &= \phi_A \circ (\phi_B \circ \phi_C) \\ &= \phi_{A(BC)}\end{aligned}$$

corresponding to the general associativity $(AB)C = A(BC)$. Indeed, we can just write

$$\phi_{ABC} = \phi_A \circ \phi_B \circ \phi_C$$

without brackets on either side.

Remark 4.18. Proposition 4.17(3) shows that, if A is an invertible matrix, then ϕ_A is an invertible map, i.e. a bijection. In fact, the converse is also true, but we shan't prove this until Theorem 4.53. In addition, we'll see in Proposition 4.50 that for a square matrix A, having a one-sided inverse is in fact enough to make A invertible.

Finally, we see that all linear maps come from matrices.

Theorem 4.19 (Linear maps come from matrices). *For any field $\mathbb{F}$, every linear map $\psi\colon \mathbb{F}^n \to \mathbb{F}^m$ is given by $\psi = \phi_A$, for a unique matrix $A \in \mathrm{M}_{m\times n}(\mathbb{F})$.*

Proof. Consider the elementary vectors $\mathbf{e}_k = \mathsf{col}_k(\mathbb{I}_n) \in \mathbb{F}^n$, that is,

$$\mathbf{e}_1 = (1, 0, \ldots, 0), \quad \ldots, \quad \mathbf{e}_n = (0, \ldots, 0, 1).$$

Every $\mathbf{x} = (x_1, \ldots, x_n)$ is a linear combination $\mathbf{x} = x_1\mathbf{e}_1 + \ldots + x_n\mathbf{e}_n$. Since ψ is linear,

$$\psi(\mathbf{x}) = x_1\psi(\mathbf{e}_1) + \ldots + x_n\psi(\mathbf{e}_n)$$

which can be written as $A\mathbf{x}$, for $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ with $\mathsf{col}_k(A) = \psi(\mathbf{e}_k) \in \mathbb{F}^m$. As the columns of A are explicitly determined by ψ, it is the unique matrix such that $\psi = \phi_A$. $\square$

4.3 Solving linear systems by Gaussian elimination

Now let's use matrices to solve systems of linear equations.

Example 4.20. The matrix equation

$$\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

is a useful shorthand for the following system of linear equations.

$$\begin{array}{ll} (1) \qquad & 3x - y = 1 \\ (2) \qquad & -x + 2y = 3 \end{array}$$

We can solve the system as follows

$$(1) + 3 \cdot (2) \quad \text{gives} \quad 5y = 10, \text{ so } y = 2.$$

Then substituting back into (1) gives

$$3x = 1 + y = 3, \text{ so } x = 1.$$

We can think of the first step here as multiplication by an 'elementary' matrix

$$\begin{pmatrix} 1 & 0 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

which simplifies the matrix to one with zeros below the diagonal, namely

$$\begin{pmatrix} 3 & -1 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 10 \end{pmatrix}.$$

As we saw above, this linear system can then be solved recursively: first solve for y, and use that to solve for x.

There are two useful points-of-view on the problem that is being solved here:

1. Writing $\mathbf{b}$ as a linear combination of the columns of A.

$$x \begin{pmatrix} 3 \\ -1 \end{pmatrix} + y \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

2. Finding the intersection of two lines L_1 and L_2 in the plane

$$\begin{array}{lll} L_1 & \text{has equation} \qquad & 3x - y = 1 \\ L_2 & \text{has equation} \qquad & -x + 2y = 3 \end{array}$$

Example 4.21. Find the line of intersection of the following two planes in $\mathbb{R}^3$,

1. $2x + 6y + 3z = 8$

2. $3x + 2y + z = 5$

We solve this as a matrix equation

$$\begin{pmatrix} 2 & 6 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ 5 \end{pmatrix}$$

Apply an elementary row operation (see Theorem 4.26) to the augmented matrix.

$$\left(\begin{array}{ccc|c} 2 & 6 & 3 & 8 \\ 3 & 2 & 1 & 5 \end{array}\right) \xrightarrow{R_2 - \frac{3}{2}R_1} \left(\begin{array}{ccc|c} 2 & 6 & 3 & 8 \\ 0 & -7 & -\frac{7}{2} & -7 \end{array}\right) \xrightarrow{-\frac{1}{7}R_2} \left(\begin{array}{ccc|c} 2 & 6 & 3 & 8 \\ 0 & 1 & \frac{1}{2} & 1 \end{array}\right).$$

In other words, the system is equivalent to

$$\begin{aligned} 2x + 6y + 3z &= 8 \\ y + \tfrac{1}{2}z &= 1 \end{aligned}$$

We could divide the top row by 2, but why introduce more fractions than needed? Choose

$$z = 2\lambda, \text{ so } y = 1 - \lambda, \text{ and hence } 2x = 8 - 6(1-\lambda) - 6\lambda = 2, i.e.\ x = 1.$$

Thus

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}$$

showing that the set of solutions is of the form $L = \{\mathbf{v} + \lambda\mathbf{w} \mid \lambda \in \mathbb{R}\}$, which is the line through $\mathbf{v} = (1, 1, 0)$ in direction $\mathbf{w} = (0, -1, 2)$.

Example 4.22. Not every system of equations has a solution. For example, the equations

1. $2x + 2y + 4z = 3$
2. $x + y + 2z = 1$

are not consistent, since $(1) - 2(2)$ gives $0 = 1$, so the system has no solutions. Geometrically, the corresponding planes are parallel and so have no intersection.

Our general goal, then, is to present a systematic way to decide when a linear system of equations $A\mathbf{x} = \mathbf{b}$ has solutions, and if so, to compute the general solution (= the set of all solutions). A general system of m linear equations in n unknowns

$$\begin{aligned} a_{11}x_1 + \ldots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \ldots + a_{mn}x_n &= b_m \end{aligned}$$

can be written in matrix form $A\mathbf{x} = \mathbf{b}$, that is,

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

It can be solved by using *elementary row operations (EROs)* to reducing A to *row echelon form (REF)* giving an equivalent system which is easier to solve. The method is called *Gaussian elimination.*

Definition 4.23 (**Row echelon form**)**.** A matrix A is in *row echelon form (REF)* if

(1) all zero rows are at the bottom,

(2) first non-zero entry in each non-zero row is 1; these are called *pivots*,

(3) the pivot in row $i+1$ is strictly to the right of the pivot in row i.

In addition, we say that A is in *reduced row echelon form* (RREF) if also

(4) all entries above each pivot are zero.

Remark 4.24. Note that parts (2) and (3) of Definition 4.23(2) imply that all entries below any pivot are zero.

Example 4.25. Here we illustrate pivots in bold:

$$\text{REF:}\quad \begin{pmatrix} \mathbf{1} & 2 & 5 & 3 & 4 \\ 0 & 0 & \mathbf{1} & 2 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \qquad \text{RREF:}\quad \begin{pmatrix} \mathbf{1} & 2 & 0 & 0 & 4 \\ 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Theorem 4.26 (**Gaussian elimination algorithm**)**.** *Any matrix M can be converted to REF or RREF by a sequence of* elementary row operations *(ERO's) of three types:*

(I) $R_i \to \lambda R_i$ $(\lambda \neq 0)$*: multiply row i by non-zero scalar,*

(II) $R_i \to R_i + \lambda R_j$*: add any multiple of row j to row i,*

(III) $R_i \leftrightarrow R_j$*: swap row i and j.*

Moreover, performing these operations on an augmented matrix $(A|\mathbf{b})$ for a system of equations produces a systems of equations that is equivalent to the original system $A\mathbf{x} = \mathbf{b}$.

Proof. The following algorithm will convert any matrix M into REF:

1. Move the first non-zero entry λ in the first non-zero column to the top row, by (III);

2. Divide the top row by λ to obtain a pivot $= 1$, by (I);

3. Make all entries below this pivot zero, by (II);

4. Repeat 1-3 on the smaller matrix comprising the columns to the right of the new pivot, BUT when picking the non-zero entry λ in step 1, ignore the rows down to the new pivot and define the row below that to be the 'top row'. Keep repeating until REF is obtained.

5. To convert REF to RREF, use (II) to make all entries above each pivot zero. $\square$

Remark 4.27. We need coefficients to be in a field for 2. (or 3., when 2. is omitted) to work. In practice, it may be easier to avoid introducing fractions by omitting 2. in the algorithm, in which case the pivots are simply non-zero numbers.

Example 4.28. At each stage, we perform multiple EROs:

$$\begin{pmatrix} 1 & 1 & 2 & 0 \\ 2 & 2 & 4 & 3 \\ 1 & 2 & 2 & 2 \\ 2 & 5 & 4 & 4 \end{pmatrix} \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - R_1 \\ R_4 - 2R_1}} \begin{pmatrix} \mathbf{1} & 1 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \end{pmatrix} \xrightarrow{\substack{R_2 \leftrightarrow R_3 \\ R_4 - 3R_3}} \begin{pmatrix} \mathbf{1} & 1 & 2 & 0 \\ 0 & \mathbf{1} & 0 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

$$\xrightarrow{\substack{\frac{1}{3}R_3 \\ R_4 + \frac{2}{3}R_3}} \begin{pmatrix} \mathbf{1} & 1 & 2 & 0 \\ 0 & \mathbf{1} & 0 & 2 \\ 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\substack{R_2 - 2R_3 \\ R_1 - R_2}} \begin{pmatrix} \mathbf{1} & 0 & 2 & -2 \\ 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 + 2R_3} \begin{pmatrix} \mathbf{1} & 0 & 2 & 0 \\ 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The final step takes REF to RREF.

Health Warning 4.29. When we perform several operations in one go, I've chosen here to index the rows in each ERO by the rows of the matrix immediately to the left. However, one should think of every such operation as multiplying by an elementary matrix, in which case, we really should perform the EROs one at a time. If we were to do this, say for $R_2 \leftrightarrow R_3$ in the second set of EROs above, then the next ERO should be called $R_4 - 3R_2$ because by the time we perform that ERO, we will already have swapped rows 2 and 3.

Example 4.30. Solve $A\mathbf{x} = \mathbf{b}$ by row reducing the augmented matrix $(A \mid \mathbf{b})$. For example, to solve the system

$$\begin{pmatrix} 1 & 3 & 1 \\ 2 & 4 & 4 \\ -1 & 2 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 11 \end{pmatrix}$$

over the field $\mathbb{R}$, we proceed as follows.

$$\left(\begin{array}{ccc|c} 1 & 3 & 1 & 4 \\ 2 & 4 & 4 & 2 \\ -1 & 2 & -6 & 11 \end{array}\right) \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 + R_1}} \left(\begin{array}{ccc|c} 1 & 3 & 1 & 4 \\ 0 & -2 & 2 & -6 \\ 0 & 5 & -5 & 15 \end{array}\right) \xrightarrow{\substack{-\frac{1}{2}R_2 \\ R_3 + \frac{5}{2}R_2}} \left(\begin{array}{ccc|c} 1 & 3 & 1 & 4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Thus the equations simplify to

$$\begin{aligned} x + 3y + z &= 4 \\ y - z &= 3 \end{aligned}$$

which are solved recursively by

$$\begin{aligned} z &= \lambda, \\ y &= 3 + z = 3 + \lambda, \\ x &= 4 - z - 3y = -5 - 4\lambda, \\ \text{i.e. } (x, y, z) &= (-5, 3, 0) + \lambda(-4, 1, 1), \quad \text{for any } \lambda \in \mathbb{R}. \end{aligned}$$

Remarks 4.31 (Key observations). 1. Once $(A \mid \mathbf{b})$ is in REF, then we can solve the linear system $A\mathbf{x} = \mathbf{b}$ provided $b_j = 0$, whenever $\mathsf{row}_j(A) = 0$. Otherwise the system is inconsistent.

2. When $\mathsf{row}_i(A) = (0 \cdots 0\, 1 * \cdots *)$ with pivot in the kth column, we get an equation for the *pivot variable* $\mathbf{x}_k$ in terms of variables x_j for $j > k$; explicitly

$$x_k = b_i - \sum_{j>k} a_{ij} x_j.$$

Hence we can solve recursively, giving the *non-pivot variables* arbitrary values.

3. If A is in RREF, then this solution is non-recursive and simply expresses each pivot variable in terms of non-pivot variables.

Lemma 4.32 (Fundamental Lemma). *Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$. If $n > m$, then there is a non-zero solution to the linear system $A\mathbf{x} = \mathbf{0}$. In other words, a homogeneous linear system with fewer equations than unknowns always has a non-trivial solution.*

Proof. When we convert the matrix A to REF, there are at most m pivots. As $m < n$, there are non-pivot variables. Choosing non-zero values for some non-pivot variables x_j gives a non-zero solution to the linear system $A\mathbf{x} = \mathbf{0}$. $\square$

Corollary 4.33. *Suppose that $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ and $X \in \mathrm{M}_{n\times m}(\mathbb{F})$ satisfy $AX = \mathbb{I}_m$ and $XA = \mathbb{I}_n$. Then $m = n$. In other words, a non-square matrix cannot be invertible.*

Proof. By Proposition 4.17, $\phi_A \colon \mathbb{F}^n \to \mathbb{F}^m$ and $\phi_X \colon \mathbb{F}^m \to \mathbb{F}^n$ are mutually inverse bijections. If $n > m$, then the Fundamental Lemma says that $A\mathbf{x} = \mathbf{0}$ has a non-zero solution $\mathbf{x} = \mathbf{v}$, as well as the solution $\mathbf{x} = \mathbf{0}$, so that ϕ_A is not injective. If $m > n$, then similarly ϕ_X is not injective. So we must have $m = n$, as claimed. $\square$

4.4 The null space and the rank-nullity theorem

Definition 4.34 (Linear subspace). A subset $V \subseteq \mathbb{F}^n$ is a *linear subspace* if

- $\mathbf{0} \in V$ (so V is non-empty); and
- $\forall\, \mathbf{v}, \mathbf{w} \in V$, $\lambda, \mu \in \mathbb{F}$, we have $\lambda\mathbf{v} + \mu\mathbf{w} \in V$.

Remark 4.35. It follows that if $\mathbf{v}_i \in V$ and $\lambda_i \in \mathbb{F}$, then $\sum_{i=1}^{k} \lambda_i \mathbf{v}_i \in V$. Geometric examples are given by lines through $\mathbf{0}$ in $\mathbb{R}^2$ and lines and planes through $\mathbf{0}$ in $\mathbb{R}^3$.

Definition 4.36 (The null space of matrix). For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, the *null space* is

$$N_A = \{\mathbf{x} \in \mathbb{F}^n \mid A\mathbf{x} = \mathbf{0}\},$$

i.e. the set of solutions to the homogeneous system.

Proposition 4.37. *The null space N_A is a linear subspace of $\mathbb{F}^n$.*

Proof. Notice that $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in N_A$. Furthermore, if $\mathbf{v}, \mathbf{w} \in N_A$ and $\lambda, \mu \in \mathbb{F}$, then

$$A(\lambda\mathbf{v} + \mu\mathbf{w}) = \lambda A\mathbf{v} + \mu A\mathbf{w} = \lambda\mathbf{0} + \mu\mathbf{0} = \mathbf{0},$$

so $\lambda\mathbf{v} + \mu\mathbf{w} \in N_A$ as required. $\square$

Definition 4.38 (Basis of a linear subspace). A *basis* of a linear subspace $V \subseteq \mathbb{F}^n$ is a list of vectors $\mathbf{w}_1, \ldots, \mathbf{w}_k \in V$ for some $k \geq 0$ such that every $\mathbf{v} \in V$ can be written *uniquely* as a linear combination

$$\mathbf{v} = \sum_{i=1}^{k} \lambda_i \mathbf{w}_i$$

with $\lambda_1, \ldots \lambda_k \in \mathbb{F}$.

Example 4.39 (The standard basis). Consider the list $\mathbf{e}_1, \ldots, \mathbf{e}_n \in \mathbb{F}^n$ of elementary vectors $\mathbf{e}_k = \mathsf{col}_k(\mathbb{I}_n)$. Then any $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{F}^n$ can be written uniquely as a linear combination

$$\mathbf{x} = \sum_{i=1}^{n} x_i \mathbf{e}_i$$

Thus $\mathbf{e}_1, \ldots, \mathbf{e}_n$ is a basis for $\mathbb{F}^n$, called the *standard basis*.

Example 4.40. Gaussian elimination finds a basis of N_A, for any matrix A. For example:

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 4 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix} \xrightarrow{R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & 1 & 2 \end{pmatrix} \quad \text{RREF}$$

The general solution of $A\mathbf{x} = \mathbf{0}$ is given by assigning arbitrary values

$$x_4 = \mu, \qquad x_3 = \lambda$$

to the two non-pivot variables, and then solving for the pivot variables

$$\begin{aligned} x_2 + x_3 + 2x_4 = 0 \qquad &\Rightarrow \qquad x_2 = -\lambda - 2\mu \\ x_1 - 2x_3 - 3x_4 = 0 \qquad &\Rightarrow \qquad x_1 = 2\lambda + 3\mu, \end{aligned}$$

i.e.

$$\mathbf{x} = \begin{pmatrix} 2\lambda + 3\mu \\ -\lambda - 2\mu \\ \lambda \\ \mu \end{pmatrix} = \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -2 \\ 0 \\ 1 \end{pmatrix}.$$

Since λ, μ are uniquely determined by $\mathbf{x}$, we see that

$$\mathbf{w}_1 = \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{w}_2 = \begin{pmatrix} 3 \\ -2 \\ 0 \\ 1 \end{pmatrix}$$

is a basis of N_A in this case.

Lemma 4.41. *If $\mathbf{v}$ is one solution to a linear system $A\mathbf{x} = \mathbf{b}$, then the general solution is*

$$\mathbf{x} = \mathbf{v} + \lambda_1 \mathbf{w}_1 + \ldots + \lambda_k \mathbf{w}_k$$

for $\lambda_1, \ldots, \lambda_k \in \mathbb{F}$ and where $\mathbf{w}_1, \ldots, \mathbf{w}_k$ is a basis of N_A.

Remark 4.42. This reads as: *The general solution to $A\mathbf{x} = \mathbf{b}$ equals a particular solution plus the general solution to the corresponding homogeneous equation $A\mathbf{x} = \mathbf{0}$*'.

Proof. To see this, note that the given vector $\mathbf{x}$ is indeed a solution, because

$$A\mathbf{x} = A(\mathbf{v} + \lambda_1 \mathbf{w}_1 + \ldots + \lambda_k \mathbf{w}_k) = A\mathbf{v} + A(\lambda_1 \mathbf{w}_1 + \ldots + \lambda_k \mathbf{w}_k) = \mathbf{b} + \mathbf{0} = \mathbf{b},$$

because $\lambda_1 \mathbf{w}_1 + \ldots + \lambda_k \mathbf{w}_k \in N_A$. Conversely, to see that any solution takes this form, let $\mathbf{x}$ be an arbitrary solution to the system $A\mathbf{x} = \mathbf{b}$. Since $A\mathbf{v} = \mathbf{b}$, we have that

$$A(\mathbf{x} - \mathbf{v}) = A\mathbf{x} - A\mathbf{v} = \mathbf{b} - \mathbf{b} = \mathbf{0},$$

so $\mathbf{x} - \mathbf{v} \in N_A$. Since $\mathbf{w}_1, \ldots, \mathbf{w}_k$ is a basis, there exists unique $\lambda_1, \ldots, \lambda_k \in \mathbb{F}$ such that $\mathbf{x} - \mathbf{v} = \lambda_1 \mathbf{w}_1 + \ldots + \lambda_k \mathbf{w}_k$. Thus, the arbitrary solution $\mathbf{x}$ and hence can be written in the given form. □

Definition 4.43 (Rank and nullity of a matrix). For any matrix $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, the *nullity* of A, denoted $\mathrm{nullity}(A)$ is the number of columns in the row echelon form of A that don't contain a pivot, while the *rank* of A is the number of columns in the row echelon form of A that do contain a pivot.

Theorem 4.44 (Rank-Nullity Theorem for matrices). *Let $A \in \mathrm{M}_{m\times n}(\mathbb{F})$. Then*

$$n = \mathrm{rank}(A) + \mathrm{nullity}(A).$$

Proof. This follows from the fact that there are n columns of A, and each either contains a pivot or it does not, giving

$$(\text{number of columns}) = (\text{number of pivots}) + (\text{number of non-pivot columns})$$

as required. □

Remark 4.45. (Nonexaminable) A linear subspace $V \subseteq \mathbb{F}^n$ can have many different bases, but it will be proved in Algebra 1B that the number of elements in a basis depends only on V (when this number is finite). This number is called the *dimension* of V, denoted $\dim V$. Thus, the dimension of the null space N_A is the nullity of A. You will also show in Algebra 1B that the dimension of the image of the map ϕ_A is the rank of A.

4.5 Invertible matrices and EROs

We now investigate why Gaussian elimination works. The first step is to establish a strong link between EROs and certain invertible matrices.

Lemma 4.46. *If $P \in \mathrm{M}_m(\mathbb{F})$ is invertible and $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, then*

$$A\mathbf{x} = \mathbf{b} \quad \iff \quad (PA)\mathbf{x} = P\mathbf{b},$$

i.e. the two linear systems have the same solutions.

Proof. First, if $A\mathbf{x} = \mathbf{b}$, then $(PA)\mathbf{x} = P(A\mathbf{x}) = P\mathbf{b}$ as required. Conversely, since P^{-1} exists, then multiplying the equation $(PA)\mathbf{x} = P$ on the left by P^{-1} gives

$$P^{-1}PA\mathbf{x} = P^{-1}P\mathbf{b}.$$

Since $P^{-1}P = \mathbb{I}_n$, this is equivalent to $A\mathbf{x} = \mathbf{b}$. $\square$

We claim that applying an ERO to a matrix A is the same as multiplying A from the left by a matrix P associated to the ERO. More precisely, we have

$$\mathsf{row}_i(PA) = \mathsf{row}_i(P)A = \sum_{j=1}^{m} p_{ij}\mathsf{row}_j(A)$$

This matrix P may be obtained by applying the ERO to the identity matrix.

The examples below are in $\mathrm{M}_3(\mathbb{F})$, but similar matrices exist in $\mathrm{M}_m(\mathbb{F})$ for any m.

(I)

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad P^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \lambda^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_2 \mapsto \lambda R_2 \qquad R_2 \mapsto \lambda^{-1} R_2$$

(II)

$$P = \begin{pmatrix} 1 & 0 & \lambda \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad P^{-1} = \begin{pmatrix} 1 & 0 & -\lambda \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_1 \mapsto R_1 + \lambda R_3 \qquad R_1 \mapsto R_1 - \lambda R_3$$

(III)

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad P^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_1 \longleftrightarrow R_2 \qquad\qquad R_1 \longleftrightarrow R_2$$

Remark 4.47. Gaussian elimination works by reducing a system of equations to an equivalent system in REF. Applying EROs gives an equivalent system because they convert $A\mathbf{x} = \mathbf{b}$ into

$$PA\mathbf{x} = P\mathbf{b},$$

where $P = P_k \cdots P_1$ is the product of the matrices of the ERO's that reduce A to REF.

These same ideas allow us to compute the inverse of a square matrix (if it exists). Indeed, when we convert any matrix A to REF or RREF, we can keep track of the P that represents the composition of the EROs by applying the EROs to the augmented matrix $(A \mid \mathbb{I}_n)$: the result will be $(PA \mid P)$, so P can be read off from the right-hand side of this augmented matrix.

Algorithm 4.48 (Gauss-Jordan method for computing the inverse of a matrix). Use EROs to convert the augmented matrix $(A \mid \mathbb{I}_n)$ to $(PA \mid P)$, where the matrix PA is in RREF. Then either:

- there is a pivot in every row of PA, and hence in every column, because A is square. That is, $PA = \mathbb{I}_n$. As P is invertible, we deduce that $A = P^{-1}$ by Lemma 4.7, and hence A is invertible with inverse $P = A^{-1}$ by Lemma 4.8(1); or

- PA has a zero row. In this case, PA is not invertible by Lemma 4.49 below, and hence, as P is invertible, A is not invertible (Lemma 4.8(2)).

Thus, we determine whether or not a square matrix A is invertible, and in the process, we compute A^{-1} when it exists.

Lemma 4.49. *If $A \in \mathrm{M}_n(\mathbb{F})$ has a zero row, then A is not invertible.*

Proof. If A has a zero row, then so does AB for any matrix B, so in particular AB can not be $\mathbb{I}_n$. □

Proposition 4.50. *If $A, X \in \mathrm{M}_n(\mathbb{F})$ and $AX = \mathbb{I}_n$, then both A and X are invertible, and in fact $X = A^{-1}$ and $A = X^{-1}$.*

Proof. By Gaussian elimination, we can find an invertible matrix P so that PA is in RREF. As observed above, either $PA = \mathbb{I}_n$, so A is invertible, with inverse $P = X$, or PA has a zero row. But the second option would mean that $PAX = P$ has a zero row, which is not possible as P is invertible. □

Example 4.51. To find the inverse of $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$:

$$\left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 0 \end{array}\right)$$
$$\xrightarrow{R_2 - 2R_1} \left(\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array}\right)$$
$$\xrightarrow{-R_2} \left(\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array}\right)$$
$$\xrightarrow{R_1 - 2R_2} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 0 & 1 & -1 & 2 \end{array}\right)$$

so

$$A^{-1} = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}.$$

You should always check (on one-side by Proposition 4.50) that we are correct:

$$\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Remark 4.52. In this case, we can also use the formula from Example 4.10

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

to write down the inverse, but the Gauss-Jordan method is the more effective method for larger matrices, even though (more complicated) formulae exist for those too.

Theorem 4.53. *If $A \in \mathrm{M}_n(\mathbb{F})$, with associated linear map $\phi_A \colon \mathbb{F}^n \to \mathbb{F}^n \colon \mathbf{x} \mapsto A\mathbf{x}$, then the following are equivalent:*

(1) The matrix A is invertible.

(2) The map ϕ_A is a bijection.

(3) The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution, for all $\mathbf{b} \in \mathbb{F}^n$.

(4) The columns of A are a basis of $\mathbb{F}^n$.

Proof. We describe the equivalence (2) $\iff$ (3) in two parts: the statement 'for all $\mathbf{b}$' in (3) is equivalent to saying that ϕ_A is surjective; while 'a unique solution' in (3) is equivalent to saying that ϕ_A is injective. The fact that (2) and (3) are equivalent follows because 'injective and surjective' means bijective.

Furthermore, for the equivalence (3) $\iff$ (4), note that

$$A\mathbf{x} = \mathbf{b} \quad \iff \quad \sum_{i=1}^{n} x_i \mathsf{col}_i(A) = \mathbf{b}.$$

Hence the left-hand side always has a unique solution iff every $\mathbf{b}$ is uniquely a linear combination of the columns of A, i.e. the columns of A form a basis (Definition 4.38).

It remains to show that (1) $\iff$ (2). The fact that (1) $\implies$ (2) is Proposition 4.17(3), so we consider the hardest part, that (2) $\implies$ (1). Suppose that A is not invertible. By Gaussian Elimination, we can find an invertible matrix P so that PA is in REF and has a zero row at the bottom of the matrix. Thus $\mathsf{row}_n(PA) = \mathbf{0}$. Hence, for any $\mathbf{b} \in \mathbb{F}^n$ with $b_n \neq 0$, there is no solution to $PA\mathbf{x} = \mathbf{b}$, that is, ϕ_{PA} is not surjective, and thus not bijective. However, $\phi_{PA} = \phi_P \circ \phi_A$ by Proposition 4.17(2) and ϕ_P is bijective, as P is invertible. Hence ϕ_A is not bijective, by Proposition 2.45(3), as required. □

4.6 Orthogonal matrices

Definition 4.54 (**Transpose**)**.** For any $A = (a_{ij}) \in \mathrm{M}_{m\times n}(\mathbb{F})$, the *transpose of* A, denoted $A^T \in \mathrm{M}_{n\times m}(\mathbb{F})$, is the matrix whose ith row is equal to the ith column of A.

Example 4.55.

$$\begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix}^T = \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix}$$

One can readily check that

$$(A^T)^T = A, \quad (\lambda A + \mu B)^T = \lambda A^T + \mu B^T, \quad (AC)^T = C^T A^T$$

for suitably sized A, B, C. For the third equality, we observe that

$$(AC)^T_{ik} = (AC)_{ki} = \sum_j (A)_{kj}(C)_{ji} = \sum_j (C^T)_{ij}(A^T)_{jk}$$

Remark 4.56. Given column vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2 = \mathrm{M}_{2\times 1}(\mathbb{R})$, notice that

$$\mathbf{v}^T\mathbf{w} = v_1w_1 + v_2w_2 = \mathbf{v}\cdot\mathbf{w} \quad \text{(scalar product)}.$$

In particular, $\mathbf{v}\cdot\mathbf{v} = \mathbf{v}^T\mathbf{v} = v_1^2 + v_2^2$ is a positive real number, when we work over $\mathbb{R}$.

Definition 4.57 (**Length, distance and angle**)**.** For any vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, define:

- the *scalar* (or *dot*) *product* is $\mathbf{v}\cdot\mathbf{w} = \mathbf{v}^T\mathbf{w}$.
- the *Euclidean length* $|\mathbf{v}|$ of a vector by $|\mathbf{v}|^2 = \mathbf{v}\cdot\mathbf{v}$, and more generally, the Euclidean distance from $\mathbf{v}$ to $\mathbf{w}$ is $|\mathbf{v} - \mathbf{w}|$.
- the *angle* $\theta \in [0, \pi]$ between nonzero vectors $\mathbf{v}, \mathbf{w}$ is defined by $\mathbf{v}\cdot\mathbf{w} = |\mathbf{v}||\mathbf{w}|\cos\theta$.

Definition 4.58 (**Orthogonal matrix**)**.** We say $A \in \mathrm{M}_n(\mathbb{R})$ is *orthogonal* if $A^TA = \mathbb{I}_n$.

Remark 4.59. By Proposition 4.50, this means that A is invertible and $A^{-1} = A^T$. By Theorem 4.53, we could also say that the columns of A form a basis and, in this case, the formula $A^T A = \mathbb{I}_n$ can be interpreted as saying that it is an *orthonormal* basis, that is, if $\mathbf{v}_i = \mathsf{col}_i(A)$, then $\mathbf{v}_i \cdot \mathbf{v}_j = 1$, if $i = j$, and $= 0$ if $i \neq j$.

Lemma 4.60. *If $A \in \mathrm{M}_n(\mathbb{R})$ is orthogonal, then $\phi_A \colon \mathbb{R}^n \to \mathbb{R}^n$ preserves lengths and angles, that is, $|A\mathbf{v}| = |\mathbf{v}|$ and more generally, $(A\mathbf{v}) \cdot (A\mathbf{w}) = \mathbf{v} \cdot \mathbf{w}$.*

Proof. Write $\mathbf{v} \cdot \mathbf{w}$ as $\mathbf{v}^T\mathbf{w}$ and so

$$(A\mathbf{v}) \cdot (A\mathbf{w}) = (A\mathbf{v})^T A\mathbf{w} = \mathbf{v}^T A^T A\mathbf{w} = \mathbf{v}^T I\mathbf{w} = \mathbf{v}^T\mathbf{w} = \mathbf{v} \cdot \mathbf{w}.$$

In particular,

$$|A\mathbf{v}|^2 = (A\mathbf{v}) \cdot (A\mathbf{v}) = \mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2.$$

□

Example 4.61. Which matrices $A \in \mathrm{M}_2(\mathbb{R})$ are orthogonal? If

$$A = \begin{pmatrix} v_1 & w_1 \\ v_2 & w_2 \end{pmatrix}, \quad \text{then} \quad A^T A = \begin{pmatrix} v_1 & v_2 \\ w_1 & w_2 \end{pmatrix}\begin{pmatrix} v_1 & w_1 \\ v_2 & w_2 \end{pmatrix} = \begin{pmatrix} |\mathbf{v}|^2 & \mathbf{v} \cdot \mathbf{w} \\ \mathbf{w} \cdot \mathbf{v} & |\mathbf{w}|^2 \end{pmatrix}$$

so

$$A^T A = \mathbb{I}_2 \iff |\mathbf{v}| = |\mathbf{w}| = 1 \text{ and } \mathbf{v} \cdot \mathbf{w} = 0,$$

that is, $\mathbf{v}$ and $\mathbf{w}$ are perpendicular unit vectors (i.e. an orthonormal basis of $\mathbb{R}^2$). Hence we must have $\mathbf{v} = \big(\cos\theta, \sin\theta\big)$ for some angle θ and then $\mathbf{w}$ is obtained by rotating $\mathbf{v}$ through $\pm\frac{\pi}{2}$, i.e. replacing θ by $\theta \pm \frac{\pi}{2}$ to get

$$\mathbf{w} = \big(-\sin\theta, \cos\theta\big) \quad \text{or} \quad \mathbf{w}' = \big(\sin\theta, -\cos\theta\big) = -\mathbf{w}.$$

In other words,

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

which is rotation through θ (and note that $\det A = \cos^2\theta + \sin^2\theta = 1$) or

$$A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix},$$

which is reflection in line of angle $\theta/2$ with horizontal axis (and note that $\det A = -1$). We can distinguish these transformations by applying A to the unit square.

4.7 3×3 determinants

The volume of the parallelepiped spanned by $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ is given by

$$V(\mathbf{u}, \mathbf{v}, \mathbf{w}) = |\mathbf{u} \cdot \mathbf{v} \times \mathbf{w}|,$$

where the 'scalar triple product'

$$\mathbf{u} \cdot \mathbf{v} \times \mathbf{w} = u_1(v_2 w_3 - v_3 w_2) + u_2(v_3 w_1 - v_1 w_3) + u_3(v_1 w_2 - v_2 w_1).$$

As we could expect, this is actually the formula for $\det B$, where

$$B = \begin{pmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{pmatrix} \in \mathrm{M}_3(\mathbb{F}).$$

The first property to check is that

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1$$

which is the volume of the unit cube.

Then

$$A^{-1} = (\det A)^{-1} \operatorname{adj} A$$

where the *adjugate* $\operatorname{adj} A$ is given by an explicit formula using smaller determinants. The 2×2 case from Example 4.10 can be thought of in this way; if

$$A = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix},$$

then

$$\det(A) = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1 b_2 - a_2 b_1 \quad \text{and} \quad \operatorname{adj} A = \begin{pmatrix} b_2 & -b_1 \\ -a_2 & a_1 \end{pmatrix}.$$

In the 3×3 case, if

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$$

then

$$\operatorname{adj} A = \begin{pmatrix} \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} & -\begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix} & \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} \\ -\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} & \begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix} & -\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} \\ \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} & -\begin{vmatrix} a_1 & b_1 \\ a_3 & b_3 \end{vmatrix} & \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \end{pmatrix}$$

It is possible to check that $A(\operatorname{adj} A) = (\det A)\,\mathbb{I}_n = (\operatorname{adj} A)A$, which in particular gives six different formulae for $\det A$, such as expansion about the first row (or other rows):

$$\det A = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

or expansion about the first column (or other columns):

$$\det A = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}$$

which was the original scalar triple product formula $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$.

All of these formulae multiply out to the same formula

$$\begin{aligned} \det A &= a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_1b_3c_2 - a_2b_1c_3 - a_3b_2c_1 \\ &= a_1b_2c_3 + c_1a_2b_3 + b_1c_2a_3 - a_1c_2b_3 - b_1a_2c_3 - c_1b_2a_3 \end{aligned}$$

which we can recognise as a sum over all permutations in S_3. This suggests the general formula(e) for the determinant of $A = (a_{ij}) \in \mathrm{Mat}_n(\mathbb{F})$

$$\det A = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma)\ a_{1\sigma(1)} \cdots a_{n\sigma(n)} = \sum_{\tau \in S_n} \operatorname{sign}(\tau)\ a_{\tau(1)1} \cdots a_{\tau(n)n}$$

The equality of the two formulae comes from setting $\tau = \sigma^{-1}$ observing that this gives a bijection $S_n \to S_n$ with the property that $\operatorname{sign}(\tau) = \operatorname{sign}(\sigma)$.

These formulae also lead to row and column expansions for $n \times n$ determinants in terms of $(n-1) \times (n-1)$ determinants, i.e. a recursive definition of $\det A$. (See Y1, S2 Algebra for a full treatment of higher determinants.)

Remark 4.62. The equality of the two formulae also means that $\det A^T = \det A$, from which it follows that any orthogonal matrix A has $\det A = \pm 1$, once one has proved that, in general, $\det \mathbb{I}_n = 1$, which is easy, and $\det AB = (\det A)(\det B)$, which is hard.

A Counting sheep

No, this section is not designed to help you fall asleep[12]. This appendix is non-examinable.

Lemma A.1. *Let $m, n \in \mathbb{N}$, and let $f\colon \{1, \ldots, m\} \to \{1, \ldots, n\}$ be an injective function. Then $m \leq n$. Furthermore, $m = n$ if and only if f is bijective.*

Proof. We prove this by induction on $m \geq 1$. For the base case, fix $m = 1$. We have $n \geq 1$ by assumption, so the first statement holds. For the second statement, our map is $f\colon \{1\} \to \{1, \ldots, n\}$. If $n = 1$, the only possible map satisfies $f(1) = 1$, so f is a bijection. For the converse, we prove the contrapositive, i.e. we show that if $n \neq 1$, then

[12] Aye, right...

f is not bijective. Indeed, $f(1) = r$ for some $r \in \{1, \dots, n\}$, and because $n > 1$, there exists some $s \in \{1, \dots, n\} \setminus \{r\}$. Thus, s does not lie in the image of f, so f is not surjective and hence not bijective. Thus $n = 1$ if and only if f is bijective.

For the induction step, suppose the statement is true for $m = k$. We want to show that it then holds for $m = k+1$. Suppose that we have an injective map $f\colon \{1, \dots, k+1\} \to \{1, \dots, n\}$. For the first part of the statement, we want to show that $k+1 \leq n$. In order to make use of the assumption that the statement is true for $m = k$, we restrict the domain of f to $\{1, \dots, k\}$. Suppose $f(k+1) = r$. Define $g\colon \{1, \dots, n\} \setminus \{r\} \to \{1, \dots, n-1\}$ by

$$g(1) = 1, \dots, g(r-1) = r-1, g(r+1) = r, g(r+2) = r+1, \dots, g(n) = n-1.$$

Notice that g is bijective. Then $g \circ (f|_{\{1,\dots,k\}})\colon \{1, \dots, k\} \to \{1, \dots, n-1\}$ is injective by Proposition 2.45(1). By the induction hypothesis $k \leq n-1$ and hence $k+1 \leq n$.

For the second statement, we must show that $k+1 = n$ iff f is bijective. For this, note that $k+1 = n$ iff $k = n-1$. Consider again the map

$$g \circ (f|_{\{1,\dots,k\}})\colon \{1, \dots, k\} \to \{1, \dots, n-1\}.$$

As $k = n-1$ iff $k+1 = n$, the induction hypothesis gives that $k+1 = n$ iff $k = n-1$ iff $g \circ f|_{\{1,\dots,k\}}$ is bijective. As g is bijective, this holds iff $f|_{\{1,\dots,k\}} = g^{-1} \circ g \circ f|_{\{1,\dots,k\}}$ is bijective. It remains to show that this last statement holds iff f is bijective. Recall that $f(k+1) = r$ and f maps $\{1, \dots, k\}$ to $\{1, \dots, n\} \setminus \{r\}$. Notice also that $r \notin \{1, \dots, n\} \setminus \{r\}$. Thus f is injective iff $f|_{\{1,\dots,k\}}$ is injective and f is surjective if and only if $f|_{\{1,\dots,k\}}$ is surjective. Thus f is bijective iff $f|_{\{1,\dots,k\}}$ is bijective which holds (as we have seen) iff $k = n-1$, i.e. $k+1 = n$. This concludes the inductive proof. $\square$

Corollary A.2. *Let A be a set and suppose there are bijective maps $f\colon \{1, \dots, m\} \to A$ and $g\colon \{1, \dots, n\} \to A$. Then $n = m$.*

Proof. The map $g^{-1} \circ f\colon \{1, \dots, m\} \to \{1, \dots, n\}$ is a compostion of bijective maps and is therefore bijective by Proposition 2.45(3). Lemma A.1 now gives that $m = n$. $\square$

= THE END[13] =

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